Evaluation of Supervised Machine Learning Models for Prediabetes Risk Prediction Using DNA Methylation Biomarkers

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Abstract

Background

Type 2 diabetes mellitus (T2D) is among the most prevalent chronic diseases globally. Prediabetes, a transitional metabolic state, characterised by elevated glycemic levels below the clinical diagnostic threshold, represents a critical window for intervention; an estimated five to ten percent of individuals with prediabetes progress to clinical T2D annually. Recent advances in epigenetics have identified DNA methylation at CpG sites as a promising biomarker class for disease risk prediction. A CpG site (Cytosine phosphate Guanine Dinucleotide) is a location in the genome where a cytosine nucleotide is followed by a guanine nucleotide in the 5' to 3' direction, linked by a phosphodiester bond. These sites are the principal targets of DNA methylation in mammals and offer predictive value. However, the majority of existing research has focused on T2D diagnosis rather than on earlier-stage prediabetes risk classification. This study seeks to address this gap.

Aims

This study aimed to evaluate the performance of four supervised machine learning algorithms: Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost), for classifying prediabetes risk using genome-wide DNA methylation data, and to assess whether methylation profiles contain a generalisable predictive signal for prediabetes risk stratification.

Methods

Genome-wide DNA methylation data from the publicly available GSE315764 dataset (hosted by the NCBI Gene Expression Omnibus) were preprocessed using variance filtering and a multi-stage feature selection pipeline comprising LASSO regression and Random Forest importance, embedded within cross-validation to prevent data leakage. All four algorithms were trained, tuned via grid search, and evaluated on a held-out test set using ROC-AUC, accuracy, F1, Matthew's Correlation Coefficient, and low-risk recall.

Results

A critical finding was revealed during data exploration: the dataset contained a perfect confound between array platform version, study cohort, and outcome class. Models trained across both cohorts achieved apparently strong discrimination (mean ROC-AUC of 0.755) by detecting the technical signature of the array platform rather than any biologically meaningful methylation signal. Upon restriction to the Discovery cohort to eliminate this confound, held-out test-set discrimination collapsed to a mean ROC-AUC of 0.605, approaching the 0.500 chance threshold. No model reliably identified low-risk individuals, with Random Forest classifying all ten low-risk test samples as high-risk despite achieving 71.1% overall accuracy. The consistency of near-

chance performance across four algorithms of markedly different inductive bias indicated that the result reflects a property of the data rather than a limitation of any particular model. Feature attribution analysis via SHAP was consequently not pursued, as attributions derived from chance-level classifiers would describe model noise rather than candidate biomarkers.

Conclusions

These findings highlight the importance of rigorous confound detection prior to model training in epigenomic machine learning studies and demonstrate that apparent predictive performance can be substantially inflated by technical batch effects. The study contributes a methodological case study in the responsible application of machine learning to high-dimensional biological data, highlighting that the absence of a predictive signal is itself an informative result when reached through sound analytical practice. Future work should employ larger, single-platform cohorts and consider multi-omics integration to determine whether a methylation-based signal for prediabetes risk exists at effect sizes below the detection threshold of the present study.

Chapter 1: Introduction

1.1 Context and Genomics

Diabetes mellitus represents one of the most significant public health challenges of the twenty-first century. An estimated 415 million people globally are living with the disease, of which approximately ninety percent have type 2 diabetes mellitus (T2D), a chronic metabolic condition characterised by persistent hyperglycaemia resulting from defects in insulin secretion, action, or both. This insulin deficiency stems from pancreatic β-cell dysfunction (Chatterjee et al., 2017; Buysschaert et al., 2011). The growing global prevalence of T2D reflects the convergence of rising obesity rates, sedentary lifestyles, ageing populations, and urbanisation, placing unprecedented pressure on healthcare systems across both developed and developing nations.

The unwanted effects of T2D extend well beyond its metabolic consequences. The World Health Organisation (WHO) has identified diabetes as a leading cause of blindness, kidney failure, heart attacks, strokes, and lower limb amputation (WHO, 2024). In the United Kingdom, approximately 3.6 million people had been diagnosed with diabetes by 2021, with a further estimate of 1 million, living undiagnosed, with T2D, unaware of the condition and its progressive impacts. The National Health Service (NHS) spends at least ten billion pounds annually on diabetes-related care, representing approximately 10 percent of its entire budget devoted to managing a single disease (Whicher et al., 2020). In 2003 and 2004, 17.1% of all US children and adolescents were classified as obese. This prevalence has continued to rise (Ogden et al., 2006).

These statistics underscore the urgent need for more effective and accurate early detection strategies. Many individuals who eventually develop clinical T2D first pass through a transitory metabolic state known as prediabetes, in which glycaemic levels are elevated above normal but have not yet exceeded the diagnostic threshold for diabetes. Prediabetes is not a benign condition, it is estimated that 5 to 10 percent of individuals with prediabetes progress to clinical T2D each year (Tabak et al., 2012). Crucially, however, this progression is not inevitable. Early identification of individuals at high risk of prediabetes enables timely lifestyle modifications and pharmacological interventions that can delay or prevent the onset of clinical disease before irreversible organ damage occurs (Adua et al., 2021). The challenge, therefore, is not merely detecting diabetes once it has developed, but identifying those individuals whose metabolic trajectory places them at elevated risk while there remains a meaningful window for intervention.

Recent advances in the field of epigenetics have opened promising avenues for addressing this challenge. DNA methylation, a fundamental epigenetic mechanism involving the addition of methyl groups to cytosine bases at CpG dinucleotide sites, plays a central role in regulating gene expression without altering the underlying DNA sequence (Kim, 2019). Importantly, DNA methylation patterns are not static, they are dynamically influenced by both genetic predisposition and environmental factors including diet, physical activity, stress, and toxicant exposure (Barros and Offenbacher, 2009). This dual sensitivity makes methylation profiles particularly valuable as

biomarkers; they capture not only inherited risk, but also the cumulative effect of modifiable exposures relevant to metabolic disease.

Progressive research has identified differential methylation at specific CpG sites associated with insulin resistance, metabolic regulation, and glucose homeostasis (Shah et al., 2019; Nadiger et al., 2024). Multiple studies have demonstrated that methylation changes can be detected in blood samples prior to T2D onset, highlighting their potential for early, non-invasive risk prediction (Raciti et al., 2021). These findings position DNA methylation data as a candidate input for predictive modelling frameworks that move beyond retrospective disease association towards prospective risk stratification.

1.2 Research Aim

The primary aim of this study is to evaluate the performance of multiple supervised machine learning algorithms for classifying prediabetes risk using genome-wide DNA methylation data while incorporating explainability. In pursuing this aim, the study aims to assess model performance using established evaluation metrics, identify CpG sites with significant predictive value for prediabetes risk through feature selection analysis, and evaluate the interpretability of model outputs for potential biomedical research applications.

1.3 Research Objectives

- Collect, process, clean and prepare DNA methylation data for machine learning analysis.
- Apply feature selection techniques suitable for high-dimensional genomic datasets.
- Train and evaluate multiple machine learning models for prediabetes risk classification.
- Compare model performance using established evaluation metrics including accuracy, F1 score, MCC, and the area under the receiver operating characteristic curve.
- Identify biologically relevant CpG sites contributing to prediabetes risk using explainable artificial intelligence methods where model performance warrants such analysis.

Chapter 2: Literature Review

2.1 Literature Search Strategy

A systematic literature review was performed to identify studies and reports at the intersection of epigenetics and machine learning for disease prevention. Key academic databases such as PubMed, Google Scholar, IEEE Xplore, and ScienceDirect were queried to ensure comprehensive coverage across molecular biology, bioinformatics, and computer science domains, maintaining the breadth of interdisciplinary scope that characterises this research area (Abolelela et al., 2007; Booth et al., 2021).

Search terms included combinations of “DNA methylation diabetes prediction”, “machine learning for genomics classification”, “epigenetic biomarkers predicting diabetes”, and “feature selection in high dimensional biological data”. Boolean operators were employed to combine key search terms systematically. For example, queries such as “Diabetes AND Methylation AND Prediction AND Machine Learning” were used to identify studies operating at the crossover of the fields (Xiao & Watson, 2017). This strategy balanced specificity, ensuring that retrieved studies addressed the relevant combination of topics, with recall, ensuring that important work was not excluded by overly narrow search criteria.

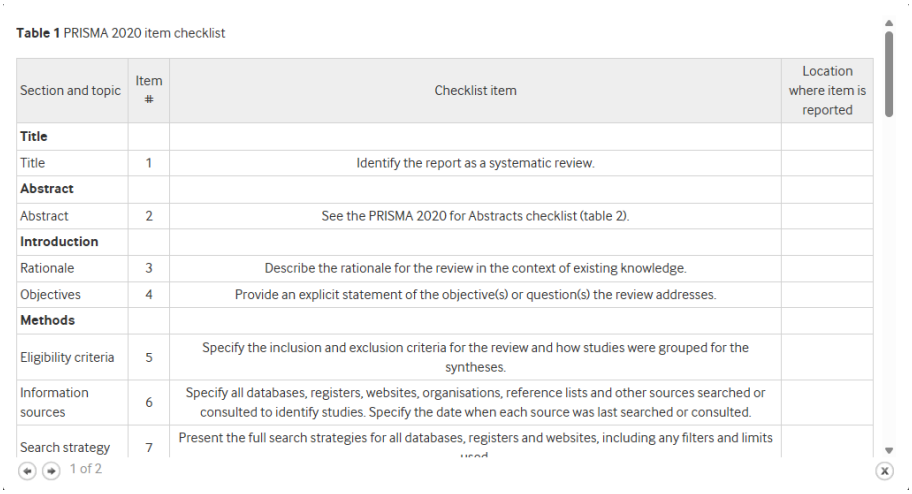


Table 1 PRISMA 2020 item checklist

Section and topic	Item #	Checklist item	Location where item is reported
Title			
Title	1	Identify the report as a systematic review.	
Abstract			
Abstract	2	See the PRISMA 2020 for Abstracts checklist (table 2).	
Introduction			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	
Methods			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits	

1 of 2

Figure 1: PRISMA 2020 item checklist for literature reviews

In line with PRISMA 2020 guidelines (Page et al., 2021), Inclusion criteria restricted the review to peer-reviewed academic studies that either applied machine learning techniques to genomic or epigenomic data in a biomedical context, or that discussed methodological principles directly relevant to such applications.

Studies were excluded if they lacked adequate validation methodology, or if they focused on applications of machine learning outside biological contexts. However, a limited number of interdisciplinary reports were included where they offered relevant theoretical insights and ideas surrounding machine learning effectiveness, that helped inform the methodological design of this study. This systematic approach aimed to guarantee the literature in use was methodologically robust, reliable, and of high quality. It made sure it was appropriate for supporting the research design and interpretation of results (Whittenmore & Knafl, 2005).

2.2 DNA Methylation and Diabetes Risk

DNA methylation is a key epigenetic mechanism that regulates gene expression and has been repeatedly, extensively linked to the onset and progression of T2D (Kim, 2019). The mechanism operates through the covalent addition of methyl groups to cytosine residues, predominantly at CpG dinucleotide sites, can silence gene transcription by altering chromatin structure or blocking transcription factor binding (Jones, 2012) (Curradi et al., 2022).

Unlike genetic mutations, methylation changes are fully reversible, and responsive to environmental stimuli, positioning them as dynamic indicators of metabolic state rather than static markers of inherited risk (Ahmed et al., 2020).

Progressive research over the past decade has identified differential methylation patterns at specific CpG sites associated with key metabolic processes including insulin resistance, glucose homeostasis and lipid metabolism (Shah et al., 2019). These epigenetic modifications are influenced by both genetic predisposition and external environmental factors such as dietary composition, physical activity levels, and exposure to metabolic stressors, making them valuable as integrative biomarkers that capture the combined effect of hereditary and lifestyle-related risk factors (Barros & Offenbacher, 2009). Nadiger et al., (2024) conducted a systematic review of DNA methylation in T2D and confirmed that specific CpG loci demonstrate consistent associations with disease status across multiple cohorts, though the effect sizes at individual sites are typically small and the biological mechanisms linking methylation changes to metabolic dysfunction remain incompletely understood.

Multiple studies have demonstrated that methylation alterations can be detected in peripheral blood samples prior to clinical T2D onset, raising the possibility that blood-based epigenetic profiling could serve as a non-invasive tool for prospective risk stratification (Raciti et al., 2021). Willmer et al. (2018) reviewed the evidence for blood-based DNA methylation biomarkers and concluded that while promising candidates exist, the translational pathway from association studies to validated clinical biomarkers remains underdeveloped.

A critical limitation identified across this body of work is that most studies focus on identifying statistical associations between methylation biomarkers and disease outcomes, rather than developing predictive models that classify individuals according to their risk of developing T2D. (Willmer et al., 2018; Ahsan et al., 2017). This emphasis on association over prediction reduces the translational utility of findings in clinical practice, where the actionable question is not whether a CpG site correlates with disease, but whether a combination of CpG sites can prospectively identify individuals who would benefit from early intervention. Yousefi et al., (2022) argue explicitly that integrating DNA methylation data into predictive modelling represents a vital and necessary step for the field.

2.3 The Use of Machine Learning in Genomic Data

The application of machine learning techniques to genomic and epigenomic datasets presents both substantial opportunities, and tough, methodological challenges. DNA methylation datasets generated by modern array platforms, such as the Illumina Infinium MethylationEPIC, routinely measure methylation levels at hundreds of CpG sites per individual (Sharma et al., 2026). A typical dataset may contain over 850,000 features per sample, while the number of available samples rarely exceeds a few hundred. This creates the well-documented high-dimensionality problem (p >>n) where the number of predictor variables vastly exceeds the number of observations. In such settings, machine learning models face a severe risk of overfitting: they may learn patterns specific to idiosyncrasies of the training data that do not generalise to unseen samples (Subramanian & Simon, 2013).

To mitigate this risk, feature selection strategies are employed to reduce the dimensionality of the feature space, whilst still retaining the features most informative for the classification task. Several complementary approaches are available. LASSO regression, introduced by Tibshirani (1996), performs embedded feature selection through L1 regularisation, which penalises the absolute magnitude of model coefficients and drives uninformative coefficients to exactly zero, effectively removing the corresponding features. Variance-based filtering provides a computationally inexpensive preliminary reduction by removing features that show minimal variation across samples and therefore possess little discriminatory power for any classification task (Zhuang et al., 2012; Meng et al., 2010). Random Forest feature importance scoring ranks features according to their contributions to classification accuracy across an ensemble of decision trees, capturing non-linear relationships that linear methods may miss (Alam et al., 2019).

Doherty et al. (2023) demonstrated that the choice of feature selection methods significantly affects the predictive performance in methylation-based models, expressing the value of evaluating multiple selection strategies rather than relying on a single approach. The rationale is that different selection methods operate on different principles and therefore identify different subsets of relevant features: L1 regularisation tends to select a sparse subset from groups of correlated features (Zou & Hastie, 2005), while tree-based importance measures may retain multiple correlated features if each independently contributes to classification. Their sequential application progressively refines the feature set from different analytical perspectives. Although principal component analysis (PCA) was considered for dimensionality reduction, the synthesis of original features into latent components would have reduced the biological interpretability of individual CpG sites, rendering the results unsuitable for biomarker identification and making its implementation inappropriate in this context (Picard et al., 2021).

2.4 Machine Learning Models for Disease Prediction

Machine Learning Models are widely applied in genomic and epigenetic research for disease classification tasks, with different algorithms offering distinct trade-offs between interpretability, predictive performance, and robustness to the specific challenges of biological data. Four algorithm types are particularly predominant in the literature and were selected for evaluation in this study.

Logistic Regression is commonly employed as a baseline model in biomedical classification due to its interpretability and its production of probabilistic outputs that facilitate clinical decision-making. Its coefficients map directly onto individual features, enabling straightforward identification of the most influential predictors (Kanduri et al., 2022). However, as a linear model, it assumes that the decision boundary between classes can be described by a linear combination of features, which may not be the case in complex biological systems.

Random Forest, an ensemble learning method that aggregates predictions from hundreds of independently trained decision trees, is frequently selected for high-dimensional datasets because it can model non-linear relationships, provides built-in feature importance metrics, and demonstrates robustness to noisy predictors (Auret and Aldrich, 2012; Koelsche, 2021). Its ensemble nature provides a natural form of regularisation, as individual tree errors are averaged across the entire forest.

Support Vector Machines (SVMs) have demonstrated strong performance in genomic studies, particularly in settings where the number of features exceeds the number of samples, as they seek the maximum-margin hyperplane that separates classes in the feature space (Guyon et al., 2002; Statnikov, 2005). This maximum-margin property provides theoretical guarantees that are particularly relevant in high-dimensional settings where the risk of overfitting is acute.

Gradient boosting methods, exemplified by XGBoost, have consistently achieved high predictive performance on structured biomedical datasets by sequentially-fitting weak learners to the residual errors of previous models. XGBoost uses built-in regularisation and is robust to overfitting while capturing complex, non-linear feature interactions (Chen & Guestrin, 2016).

Comparative studies across these algorithm groups reveal that each involves trade-offs between interpretability, raw predictive performance, and computational cost. Ensemble and boosting methods tend to outperform linear models in terms of accuracy on complex datasets, but their decision processes are substantially less transparent. They often function as grey and black boxes, where interpretation is extremely difficult, or impossible to those less technical. This limits their utility in clinical settings where model outputs must be understood, and trusted by professionals (Pintelas et al., 2020; Amann et al., 2020).

Informed by these trade-offs, this study evaluates all four algorithm types including Logistic Regression, Random Forest, SVM, and XGBoost; using a wide range of models spans the full interpretability-performance spectrum. This selection enables a systematic comparison of whether any algorithmic class can extract a generalisable predictive signal from high-dimensional methylation data. Crucially, the study also incorporates model interpretability as an explicit design objective rather than an afterthought; where model performance warrants it, Shapley Additive Explanations (SHAP) will be used

to attribute predictions to individual CpG features, ensuring that model outputs remain transparent and clinically meaningful rather than functioning solely as opaque classification instruments (Linardatos et al., 2020).

2.5 Explainability in Machine Learning for Healthcare.

A growing body of literature emphasises that predictive performance alone is insufficient for clinical machine learning applications; model outputs must be interpretable to the clinicians and researchers who act upon them (Rudin, 2019; Amann et al., 2020). Explainable artificial intelligence (XAI) encompasses a range of techniques designed to make the internal decision process of complex models accessible to human understanding (Linardatos et al., 2020). Among these, SHAP, introduced by Lundberg and Lee (2017) and grounded in cooperative game theory, have become widely adopted in in biomedical machine learning because they provide mathematically consistent attributions of each features contribution to individual predictions. SHAP offers both local predictions quantifying why a specific sample received a particular classification, and global explanations, identifying which features were most influential across an entire dataset. Its model-agnostic nature enables consistent comparison of feature attributions across algorithms with different internal structures, which is particularly valuable in comparative studies such as the present one (Lundberg & Lee, 2017). The integration of SHAP into the analytical workflow of this study was therefore motivated by both the clinical imperative for transparency and the methodological requirement for cross-model comparability.

2.6 Research Gap

Despite advances in both epigenetic research and machine learning applications, several important gaps persist at their intersection. DNA methylation has been widely acknowledged as a promising biomarker class for T2D, yet the great majority of existing work focuses on disease diagnosis, examining methylation differences between individuals who already have T2D and healthy controls. Far fewer studies address the earlier and clinically more important stage of prediabetes, where early intervention is the most effective at altering disease trajectory. A recent longitudinal epigenome wide- association study identified CpG sites associated with T2D progression, but the analysis of prediabetes-specific markers was limited and the authors called for larger studies to validate their findings (Lai et al., 2025).

In addition, despite the widespread use of machine learning techniques in biomedical data analysis, there is a notable absence of systematic comparisons between different classification algorithms applied specifically to prediabetes risk stratification using methylation data. This prevents the identification of which modelling approaches are most suitable for this particular combination of data type and prediction task. The few studies that do compare multiple algorithms typically do so in the context of clinical or demographic predictors rather than high-dimensional epigenomic features (Almadhoun & Burhanuddin, 2025; Li et al., 2025), where the challenges of dimensionality, feature correlation and sample size are qualitatively different (Gao et al., 2023; Hu et al., 2022).

Furthermore, while many genomic machine learning studies prioritise accuracy and overall model performance in their reporting, they frequently neglect the importance of model interpretability, thereby reducing their clinical applicability and their ability to generate biological insights from model outputs (Ennab and Mcheik, 2024). Rudin (2019) highlights the widespread use of black box machine learning models in high-stakes contexts, including healthcare judicial contexts.

In high-risk domains such as healthcare, interpretability is essential, not merely as a desirable property but as a prerequisite for fostering trust, ensuring transparency, and enabling informed clinical decision-making (Rudin, 2019). Therefore, this study aims not only to classify individuals and compare model performance, but to do so in a reproducible and explainable manner, minimising reliance on opaque modelling techniques and prioritising analytical rigour over headline performance figures.

Chapter 3: Methodology

3.1 Structure of Methodology



Figure 2: Flow diagram showing overview of method structure and processes. Includes Data acquisition, Initial preprocessing, Confound detection, Cohort restriction and re-filtering, Stratified train/ test split, Feature selection, Model training and tuning, Model evaluation and Interpretability assessment.

Figure 2 shows an overview of the data collection, processing, machine modelling, tuning and evaluation pipeline that was used to conduct this analysis. The following sections go into further detail on the

nuances of these stages, the obstacles encountered, and the detection of a platform confound that demanded the redesign of downstream processes.

3.2 Research Design

This study used a quantitative, computational research design to evaluate the effectiveness of supervised machine learning models for classifying prediabetes risk from genome-wide DNA methylation data. The methodological goal was to build, train and compare multiple classification models using blood-based epigenetic markers as predictive features, with prediabetes risk cluster membership as the target variable.

The choice of machine learning as the analytical framework was motivated by the extremely high dimensionality of the dataset: each sample contained methylation measurements at hundreds of thousands of CpG sites, resulting in a high-dimensional and complex feature space that presents challenges for conventional analytical approaches (Zhang et al., 2019). Machine learning algorithms are specifically designed to uncover patterns and correlations among predictive features in high-dimensional settings such as this, making them the appropriate analytical tool for this data type.

The study evaluated four different supervised classification algorithms: Logistic Regression, Random Forest, Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost). This selection of models spans a range of complexities and inductive biases, from linear and fully interpretable (LR) to the non-linear, ensemble-based (Random Forest, XGBoost)/, with SVM occupying an intermediate position. By evaluating algorithms that learn in different ways, the study aimed to determine whether any approach could extract a generalisable predictive signal from the methylation data, and to assess whether model performance variances reflected genuine algorithmic advantages or were artefacts of the data structure.
The project workflow followed a standard data science pipeline consisting of data acquisition, data preprocessing, feature selection, model training, model evaluation and interpretability analysis (Biswas et al., 2022).

Critically, feature selection and feature scaling were embedded within scikit-learn pipeline objects, and executed inside cross-validation loops, ensuring that information from held-out data could not leak into the feature selection process. This design decision was essential given the extreme dimensionality of the data: with approximately 194,000 features and only 187 samples, any data leakage would have inflated performance estimates substantially and rendered the results unreliable.

3.3 Dataset

The DNA methylation dataset used in this study is GSE315764 (Singh et al., 2026), a publicly available dataset hosted by the NCBI Gene Expression Omnibus (GEO). The dataset contains genome-wide methylation measurements derived from human blood samples assayed using the Infinium Methylation EPIC platform (Pidsley et al., 2016). Methylation levels are represented as beta values ranging from zero to one, where zero indicates no methylation and one indicates complete methylation at a given CpG

site. The dataset comprised 333 human samples drawn from two study cohorts: a Discovery cohort of 187 samples, and a Replication cohort of 146 samples. The original study by Singh et al. (2026) stratified these individuals into clinically defined prediabetes clusters based on their metabolic risk profiles associated with elevated probability of T2D progression. For the purpose of this study, the original, multi-cluster classification was collapsed into a binary target variable: low risk (cluster 2, coded as 0) versus high-risk (clusters 3, 5, and 6, coded as 1). This grouping followed the stratification established by Singh et al. (2026), in which cluster 2 was characterised by a metabolically favourable profile with low insulin resistance, low hepatic fat, and preserved beta-cell function, whereas clusters 3, 5 and 6 each exhibited distinct but consistently elevated metabolic risk features including increased insulin resistance, hepatic steatosis, or impaired insulin secretion. This binary formulation was chosen to maximise the statistical power available for each class given the samples size, and because the clinical decision of interest, whether an individual requires heightened monitoring or intervention, is itself fundamentally binary.

Each sample therefore contained a high-dimensional vector of CpG methylation beta values as predictor variables and a binary risk cluster label as the target variable, supporting the direct application of supervised machine learning classification.

3.4 Data Preprocessing

Prior to model training, comprehensive preprocessing was applied to ensure data quality, structural consistency, and analytical validity. The preprocessing pipeline comprised several sequential stages, each addressing a specific requirement of the downstream machine learning analysis.

The raw methylation data file provided by GSE315764 stored CPG sites as rows and samples as columns. Since scikit-learn and all standard machine learning frameworks expect samples as rows and features as columns, the matrix was transposed to conform to this convention (Scikit-learn developers, 2026). Sample labels, including risk cluster membership, study cohort, and array platform information, were extracted from the GEO series matrix metadata files and joined to the transposed methylation matrix using sample identifiers.

An initial variance filter was applied to remove CpG sites exhibiting minimal variation across all 333 samples. A threshold of 0.001 on the variance of beta values was used, removing sites that were essentially constant across individuals and therefore incapable of contributing to any classification task. This initial filter reduced the feature space from 677,703 CpG sites to 288, 720, a 57.4% reduction that primarily eliminated technically invariant probes.

Following the identification of a platform confound in the dataset structure (explained further in section 4.1), the analysis was restricted to the 187 samples in the discovery cohort. The variance filter was then recomputed using this single-platform subset, as some CpG sites that had appeared variable across the full dataset were only so because of systematic differences between the two array platforms rather than any genuine biological variation between individuals. This re-computation removed a further 95,082 CpG sites, yielding 193,638 features for model training. As a result, one third of the previously retained

features provided mechanistic evidence for the platform confound identified through metadata inspection.

Feature scaling was performed using z-score standardisation, transforming each CpG feature to have zero mean and unit variance within the training data (Cheadle et al., 2003). This step was essential for models whose optimisation procedures are sensitive to feature magnitude, particularly LR and SVM, both of which penalise coefficient magnitudes and would therefore treat features on different scales unequally in the absence of standardisation (Choi, 2024; Mathivanan et al., 2025). This ensured that differences in feature magnitude did not disproportionately influence model optimisation. Standardisation was applied within the scikit-learn Pipeline object to ensure that scaling parameters were computed exclusively on training data and applied to test data without leakage.

3.5 Feature Selection

A multi-step feature selection strategy was implemented to reduce the dimensionality of the methylation feature space while retaining the CpG sites most informative for prediabetes risk classification. This strategy included three sequential stages, each operating on a different analytical principle, and was entirely within the cross-validation loop to prevent information leakage from held-out data into the feature selection process.

The first stage, variance filtering, was applied during preprocessing as described in Section 3.3 and served as a computationally inexpensive initial reduction that removed features with near-zero variance, which provide limited discriminatory information and may reduce statistical power when retained in high-dimensional methylation datasets (Zhuang et al., 2012; Meng et al., 2010).

The second stage employed LASSO (Least Absolute Shrinkage and Selection Operator) regression as a regularisation-based embedded feature selector. LASSO applies an L1 penalty to the magnitude of regression coefficients, driving the coefficients of uninformative features to exactly zero (Tibshirani, 1996). The scikit learn SelectFromModel wrapper was used to retain only those features assigned non-zero coefficients by a Logistic Regression model with L1 penalty, inverse regularisation strength C=0.1, and balanced class weights. This stage was particularly suited to the methylation data because CpG sites in genomic proximity are often highly correlated, and LASSO tends to select a sparse subset from such correlated groups, effectively performing implicit dimensionality reduction in the presence of multicollinearity (Zou & Hastie, 2005). Although this behaviour means that some informative features may be arbitrarily excluded in favour of correlated alternatives, this trade-off is acceptable at this stage, where the primary goal is dimensionality reduction rather than exhaustive feature identification.

The third stage applied Random Forest feature importance to rank the features surviving LASSO selection according to their mean decrease in impurity across an ensemble of 200 decision trees (Rogers & Gunn, 2006). Features scoring above the mean importance threshold were retained. This stage complemented LASSO by capturing non-linear contributions to classification that a linear model cannot detect: LASSO evaluates each feature's marginal linear association with the outcome, whereas Random Forest importance reflects a feature's contribution to reducing class impurity across varied non-linear

splits. Doherty et al. (2023) demonstrated that the choice of feature selection methods significantly affects predictive performance in methylation-based strategies rather than relying on a single approach.

The exact number of features retained after the LASSO and Random Forest selection stages varied across cross-validation folds, as both stages were refitted within each fold to prevent data leakage. The input to these stages was the 193,638 CpG sites surviving variance filtering. The LASSO stage typically reduced this to several hundred features, and the subsequent Random Forest importance stage further reduced the set to those scoring above the mean importance threshold. Because feature counts differed across folds and were determined entirely by the data within each fold, no single fixed number can be reported; this variability is an expected consequence of the leakage-safe pipeline design.

3.6 Machine Learning Model Development

Model	Parameter	Values Explored
Logistic Regression	C (inverse regularisation strength)	0.01, 0.1, 1, 10
Random Forest	n_estimators (number of trees)	200, 400
Random Forest	max_depth (maximum tree depth)	None, 5, 10
SVM	C (regularisation parameter)	0.1, 1, 10
SVM	kernel	linear, rbf
XGBoost	n_estimators (number of trees)	200, 400
XGBoost	max_depth (maximum tree depth)	3, 6
XGBoost	learning_rate	0.05, 0.1

Table 1: Hyperparameter search space for grid search cross-validation. All combinations were evaluated across five stratified folds with ROC-AUC as the optimisation criterion.

Four supervised machine learning models were developed and trained to classify individuals into prediabetes risk groups: low-risk (cluster 2) versus high-risk (clusters 3, 5 and 6). The selected algorithms widely applied in clinical studies, were Logistic Regression, Random Forest, Support Vector Machine and XGBoost (Omar et al., 2024).

All four models were embedded within scikit-learn Pipeline structures comprising a StandardScaler, a LASSO-based feature selector, a Random Forest-based feature selector, and the classifier itself.

This Pipeline architecture guaranteed that scaling parameters, feature selection boundaries, and model coefficients were fitted exclusively on training data within each cross-validation fold or grid search

iteration. As seen in Table 1, class imbalance was addressed through algorithm specific mechanisms: Logistic Regression, Random Forest, and SVM used the class_weight="balanced" parameter, which automatically up-weights the minority class in inverse proportion to its frequency; XGBoost used the analogous scale_pos_weight parameter set to the ratio of majority to minority class samples (approximately 2.7 after restriction to the Discovery cohort.)

The training data comprised 80% of the Discovery cohort samples (149 samples: 40 low risk, 109 high-risk), with the remaining 20% (38 samples: 10 low-risk, 28 high-risk) held out as a genuine unseen test set. Stratified sampling was also used to preserve the class proportion across both partitions. All random processes were seeded with a fixed value to ensure full reproducibility of results.

Hyperparameter tuning was conducted using GridSearch CV with five-fold stratified cross-validation on the training set, optimising for ROC-AUC rather than accuracy. This choice of optimisation was deliberate because accuracy can be misleading in imbalanced data sets, where a classifier can achieve a high accuracy simply by predicting the majority class. As expresses by Peters et al., (2024) "accuracy is an inappropriate measure to use for imbalanced datasets." When performance on the minority class is of interest.

Accuracy is inflated by majority class prediction in imbalanced datasets, whereas ROC-AUC measures a model's ability to discriminate between classes across all possible decision thresholds. The hyperparameter grids explored were kept deliberately small to limit the computational cost of refitting the full pipeline, including both feature selection stages, on each of the five folds for every parameter combination.

3.7 Model Evaluation

Model performance was evaluated using multiple complementary classification metrics to provide a comprehensive assessment that accounted for the class-imbalance structure of the dataset. Overall accuracy was reported as the proportion of correctly classified samples, with the caveat that accuracy can be misleading in imbalanced settings where majority class prediction achieves high scores despite offering no clear clinical utility.

The F1 score, computed as the harmonic mean of precision and recall for the high-risk class, was included to assess the imbalance between the model's ability to correctly identify high-risk individuals (recall) and the proportion of high-risk prediction that were correct (precision). However, since an often-overlooked clinical concern in prediabetes screening is identifying those at low-risk who may otherwise be over-treated or inadequately monitored, low-risk recall (the proportion of truly low-risk individuals correctly identified as low-risk) was reported separately and also prioritised in interpretation.

The area under the receiver operating characteristic curve (ROC-AUC) was calculated for each model to assess its ability to discriminated between risk classes across all possible classification thresholds. ROC-AUC ranges from 0.0 to 1.0, with 0.5 representing chance-level discrimination and 1.0 representing perfect separation of the two classes. This metric was chosen as the primary basis for model comparison

since it is threshold-independent and insensitive to class imbalance. Confusion matrices were generated for each model to visualise the distribution of true positives, true negatives, false positives and false negatives, providing a granular view of each model's classification behaviour. Five-fold stratified cross-validation was employed during both initial evaluation and hyperparameter tuning, with each fold preserving the class distribution of the full training set (Fawett., 2006).

Matthew's correlation Coefficient (MCC) was also computed for each model. MCC produces a balanced measure of classification quality that accounts for all four quadrants of the confusion matrix simultaneously, returning values between -1 (perfect misclassification) and +1 (perfect classification), with 0 indicating performance no better than random assignment. Unlike accuracy and F1, MCC is not inflated by correct majority-class predictions and is widely regarded as the most informative single metric for evaluating binary classifiers on imbalanced datasets (Chicco & Jurman, 2020).

3.8 Model Interpretability and Biomarker Identification

The study design incorporated provisions for model interpretability and biomarker identification through Shapley Additive Explanations (SHAP), a framework that quantifies the contribution of each feature to individual model predictions (Lundberg & Su-In Lee). SHAP was selected because it provides both local explanations (why specific sample was classified as it was) and global explanations (which features were most influential across the dataset), and because it is model-agnostic, enabling consistent comparison across the four algorithms evaluated.

However, the study design also incorporated an explicit precondition: feature attribution analysis would only be pursued if the models achieved discrimination meaningfully above chance on the held-out test set. Attributions derived from a model that does not reliably separate the outcome class describe the model's internal noise rather than any biological signal and reporting them as candidate biomarkers would constitute over-interpretation. This precondition reflects the principle that interpretability tools explain what a model has learned, and when a model has not learned anything generalisable, there is nothing meaningful to explain (Rudin, 2019). The decision on whether to proceed with SHAP analysis was therefore deferred until the test-set evaluation results were known.

3.9 Ethical Considerations

This study was conducted in compliance with the principles of the General Data Protection Regulation (GDPR), specifically Article 5 concerning purpose limitation and data minimalization (European Parliament and Council of the European Union, 2016). The dataset used, GSE315764, is publicly available through the NCBI Gene Expression Omnibus, a well-established scientific repository maintained by the National Centre for Biotechnology Information (Singh et al., 2026). The dataset does not contain any directly identifiable personal information: samples are identified by anonymised accession numbers and cluster labels with no names, dates of birth, or other identifying metadata available. This substantially reduces the privacy risks associated with the study.

While genomic data is inherently sensitive due to its uniqueness to individuals, the methylation beta values in GSE315764 represent aggregate methylation proportions at predefined probe sites rather than raw genetic sequence data, further limiting the potential for re-identification. The data were used solely for the purposes of this research project, and al analytical processes were documented transparently to support reproducibility. No additional data collection was conducted, and no human subjects were recruited for this study. A copy of the ethics approval documentation is provided in the Appendix.

Chapter 4: Results

4.1 Dataset Characteristics and Identification of a Platform Confound

The GSE315764 dataset contained genome-wide DNA methylation measurements for 333 human blood samples, assayed across two versions of the Illumina Infinium MethylationEPIC array platform. Each sample comprised methylation beta values, which were continuous values between zero and one representing the proportion of methylated cytosines, measured at 677,703 harmonised CpG sites. The original study classified individuals into four prediabetes risk clusters: cluster 2 (low-risk metabolic profile) and clusters 3,5 and 6 (high-risk metabolic profiles associated with elevated probability of progression to T2D). No cluster 4 was present in the dataset. For this study, the four clusters were collapsed into a binary target: cluster 2 as low-risk (n = 50) and clusters 3,5 and 6 as high-risk (n = 283). Table 2 presents the distribution of samples by cluster and study cohort.

Cluster	Risk Group	Discovery	Replication	Total
C2	Low-risk	50	0	50
C3	High-risk	42	49	91
C5	High-risk	45	51	96
C6	High-risk	50	46	96
Total		187	146	333
Low Risk		50	0	50
High risk		137	146	283

Table 2: Composition of the GSE 315764 cohort by prediabetes cluster and study cohort. (dataset contains no cluster 4) All 50 low-risk samples fall within the Discovery cohort.

Inspection of sample metadata prior to modelling revealed that cohort membership, array platform, and outcome class were perfectly confounded. Every sample in the Discovery cohort has been assayed on Infinium MethylationEPIC v1 (GPL21145), while every sample in the Replication cohort had been assayed on version 2 (GPL33022). Critically, all 50 low-risk individuals were in the Discovery cohort; the replication cohort was composed entirely of high-risk samples. This structure meant that array version predicted risk group with complete accuracy across the full dataset, not because methylation biology differed between risk groups, but because the experimental design was confounded by platform. Table 2 presents this confound explicitly.

Cohort	Array Platform	n	Low-Risk	High-Risk	% High-Risk
Discovery	EPIC v1 (GPL21145)	187	50	137	73.30%
Replication	EPIC v2 (GPL33022)	146	0	146	100.00%

Table 3: Cohort, array platform, and outcome are perfectly confounded. Every Replication sample was assayed on EPIC v2 and belongs to a high-risk cluster; consequently, array version predicts risk groups with complete accuracy across the full dataset.

This confound has a clear mechanistic basis. Successive versions of the Illumina EPIC array differ in probe content, signal intensity distributions, and technical characteristics, producing systematic differences in measure beta values that are attributable to the platform rather than to any underlying biological variation (Pidsley et al., 2016). Supervised machine learning classifiers are well documented to exploit such batch effects wherever they correlate with the outcome variable, as the technical signature provides a shortcut that is easier to learn than any genuine biological signal (Leek et al., 2010).

This confound had two direct consequences for the present study. First, the discovery-to-replication validation strategy employed by the original authors of GSE315764 could not be reproduced for a low-risk versus high-risk classification task, because the Replication cohort contains no low-risk samples, rendering the classification task undefined within that subset. Second, any model that was trained across both cohorts would risk achieving apparently strong predictive performance by detecting array version rather than any methylation signal of biological origin. Thereby producing results that would not generalise to data collected on a different platform or at a different time.

The decision was therefore made to restrict all subsequent analysis to the 187 samples of the Discovery cohort, which were assayed on a single platform (EPIC v1) and continued both risk classes. This restriction carried a substantial cost: 146 samples representing 43.8% of the total dataset, (seen in Table 3) were excluded, leaving fewer observations for model training and evaluation. However, those samples were contributing confound rather than information; retaining them would have inflated apparent model performance while undermining the validity of every downstream result. Restricting to a single platform also improved the class balance from 15.0% low-risk (50 of 333) to 26% low-risk (50 of 187), which, while still imbalanced, made the classification task more tractable.

Pipeline Stage	CpG Sites Retained	% of Original
Combined harmonised matrix (as supplied)	677,703	100.00%
After variance filter (333 samples, both platforms)	288,720	42.60%
After variance re-filter (Discovery only, single platform)	193,638	28.60%

Table 4: Feature reduction across the preprocessing pipeline.

Supporting evidence for the platform confound was provided by the variance filter re-computation. When variance filtering was applied to the full 333-sample dataset spanning both platforms, 288,720 CpG sites were retained. When the same filter was recomputed within the Discovery cohort alone (a single platform), a further 95,082 sites were removed, reducing the retained set to 193,638. Table 4 summarises this progressive feature reduction.

The loss of one-third of the previously retained CpG sites upon restriction to a single platform is itself diagnostic. Those 95,082 sites had appeared variable across the full dataset not because individuals differed at those positions, but because the two array platforms measured them differently. This observation provides independent, mechanistic corroboration of the confound identified through metadata analysis: it confirms that the platform confound was not merely a labelling coincidence, but was actively injecting variance into the feature space, variance that a machine learning model would readily exploit.

Set	n	Low-risk	High-risk	% Low-risk
Full Discovery cohort	187	50	137	26.70%
Training (80%)	149	40	109	26.80%
Held-out test (20%)	38	10	28	26.30%

Table 5: Analysis dataset following restriction to the Discovery cohort. (Held out test set retains only 10 low-risk individuals, a constraint limiting statistical precision of minority-class performance estimates).

As shown by Table 5, the analysis proceeded using the 187 Discovery cohort samples partitioned into an 80% training set (149 samples: 40 low-risk, 109 high-risk) and a 20% held out test set (38 samples: 10 low-risk, 28 high-risk), with stratified sampling preserving class proportions across both splits. Table 5 conveys the results of this process.

It is important to acknowledge what the cohort restriction did not solve. Only 50 low-risk individuals existed in the entire dataset, and after the 80/20 split, only 10 remained in the test set. Any evaluation of a model's ability to identify low-risk individuals therefore rested on these 10 cases, conferring limited

statistical power and wide implicit confidence intervals on per-class metrics. This constraint was inherent to the dataset and could not be addressed through analytical design choices; it is noted here so that subsequent results are interpreted with appropriate caution rather than appearing to have been discovered only after the results proved inconvenient.

4.2 Cross–Validated Performance on the Training Set

All four models were first evaluated under default hyperparameter settings using five-fold stratified cross-validation on the training set, with feature selection and scaling performed within each fold to prevent information leakage. Table 6 presents the cross validates performance prior to hyperparameter tuning.

Model	Accuracy	F1	ROC-AUC
Logistic Regression	0.678 ± 0.060	0.794 ± 0.043	0.661 ± 0.124
Random Forest	0.718 ± 0.033	0.833 ± 0.024	0.649 ± 0.112
SVM	0.692 ± 0.051	0.804 ± 0.037	0.683 ± 0.159
XGBoost	0.725 ± 0.013	0.838 ± 0.009	0.641 ± 0.120

Table 6: Cross validated performance prior to hyperparameter tuning (five-fold stratified cross-validation on the training set; mean ± standard deviation across folds) all feature selection was performed within each training fold.

An initial evaluation under default hyperparameter settings confirmed that all four algorithms performed similarly despite their markedly different inductive biases. Two observations were immediately apparent from these results. Firstly, the accuracy and F1 scores appeared relatively healthy, ranging from 0.678 to 0.725 and 0.794 to 0.838 respectively. However, these metrics were inflated by the class imbalance: a trivial classifier predicting high-risk for every sample would achieve approximately 73% accuracy, which exceeds the observed accuracy of two of the four models. The ROC-AUC values, which are insensitive to class distribution and measure actual discrimination ability, told a more sobering story. All four models clustered narrowly between 0.641 and 0.683, barely above the 0.5 chance threshold, and their standard deviations across folds were large (0.112 to 0.159), indicating substantial instability in fold-to-fold estimates. This variability was attributable to the small number of low-risk training samples: with only 40 low-risk cases, and the gain or loss of even one or two such cases between folds materially altered the performance estimate.

Second, all four algorithms performed similarly despite their markedly different inductive biases. Logistic Regression, a linear model, achieved an AUC of 0.661; XGBoost, a complex non-linear ensemble, achieved 0.641. The narrow spread across algorithms that learn in fundamentally different ways suggested that the limitation resided in the data rather than in the choice of model, as a signal that one algorithm class could exploit would typically produce a meaningful separation from the others.

Hyperparameter tuning was then performed using GridSearchCV with five-fold stratified cross-validation, optimising for ROC-AUC. The tuned pipeline included both LASSO and Random Forest selection stages, completing the multi-step selection strategy described in the methodology. Table 7 presents the tuning outcomes.

Model	Best CV ROC-AUC	Best CV MCC	Selected Hyperparameters
Logistic Regression	0.634	0.135	C = 10
Random Forest	0.600	0.037	max_depth = 5, n_estimators = 200
SVM	0.641	0.147	C = 10, kernel = rbf
XGBoost	0.621	0.211	learning_rate = 0.05, max_depth = 3, n_estimators = 200

Table 7: Hyperparameter tuning outcomes (grid search, five-fold stratified cross-validation, optimised for ROC-AUC). The two-stage LASSO and Random Forest feature selection was applied within each fold. MCC is shown.

Counter-intuitively, hyperparameter tuning reduced cross validated AUC across all four algorithms relative to the untuned baselines. As shown in Table 7, Logistic Regression fell from 0.661 to 0.634; SVM, the best performing model, fell from 0.683 to 0.641. This result, while unexpected, is consistent with a well-understood phenomenon in machine learning: where no parameterisation captures genuine generalisable structure in the data, grid search selects the configuration that best matches the idiosyncrasies of the particular cross-validation folds evaluated, a form of overfitting to the validation procedure itself. The addition of the second feature selection stage (Random Forest importance) further reduced the feature set, which in the absence of a strong signal served primarily to increase variance in the estimates rather than to refine them. This observation is diagnostic: it provides further evidence that

the methylation features, after confound removal, did not contain a signal strong enough for supervised classification at this sample size.

The cross-validated MCC values reinforced this interpretation. MCC ranged from 0.037 (Random Forest) to 0.211 (XGBoost), with all four models falling well below the threshold that would indicate meaningful classification quality. An MCC of 0.0 represents performance no better than random assignment, and while all four values were nominally positive, their proximity to zero confirmed that no model achieved reliable discrimination between the two risk groups during cross-validation. Notably, Random Forest produced the lowest MCC at 0.037, effectively indistinguishable from chance, despite achieving the joint-highest accuracy (0.718) among the untuned baselines. This pattern is consistent with the metrics known insensitivity to majority class prediction in imbalanced settings.

4.3 Performance on the Held-Out Test Set

Model	Accuracy	F1	ROC-AUC	MCC	Low-risk Recall	Low-risk ID
Logistic Regression	0.711	0.800	0.600	0.277	0.50	5/10
Random Forest	0.711	0.831	0.621	−0.098	0.00	0 / 10
SVM	0.711	0.807	0.646	0.229	0.40	4/10
XGBoost	0.658	0.787	0.554	−0.056	0.10	1/10

Table 8: Final performance of tuned models on the held-out test set (n=38; 10 low-risk, 28 high-risk). The test set was evaluated once, after all model selection and tuning were complete. All ROC-AUC values approach the 0.5 chance threshold.

The tuned models were evaluated once on the 38-sample held-out test set. These samples had played no part in feature selection, model fitting, or hyperparameter tuning, so these results represent an honest estimate of performance on genuinely unseen data. Table 8 summarises the test-set metrics.

The MCC values further confirmed the absence for meaningful discrimination. An MCC of 0.0 indicates classification no better than chance, and the values were consistent with near random assignment. As previously stated, this metric is particularly informative in this setting because it is not inflated by correct majority-class predictions, unlike accuracy and F1, and therefore provides a more honest summary of classification quality on an imbalanced dataset. The MCC values on the held-out test set provided the most unambiguous confirmation of classification failure. Random Forest and XGBoost produced negative MCC values of -0.098 and -0.056 respectively, indicating that their predictions were not merely uninformative, but inversely associated with the true class labels, a result worse than random assignment. Logistic Regression achieved the highest MCC at 0.277 and SVM at 0.229, both nominally positive but far below the levels that would indicate clinically useful discrimination. The contrast between these MCC values and the corresponding accuracy scores is itself instructive: Random

Forest achieved 71.1% accuracy with an MCC of -0.098, demonstrating that accuracy and MCC can not only disagree but point in opposite directions when class imbalance is preset. This divergence validates the inclusion of MCC as an evaluation metric and supports the broader conclusion that no model learned a generalisable distinction between the two risk groups.

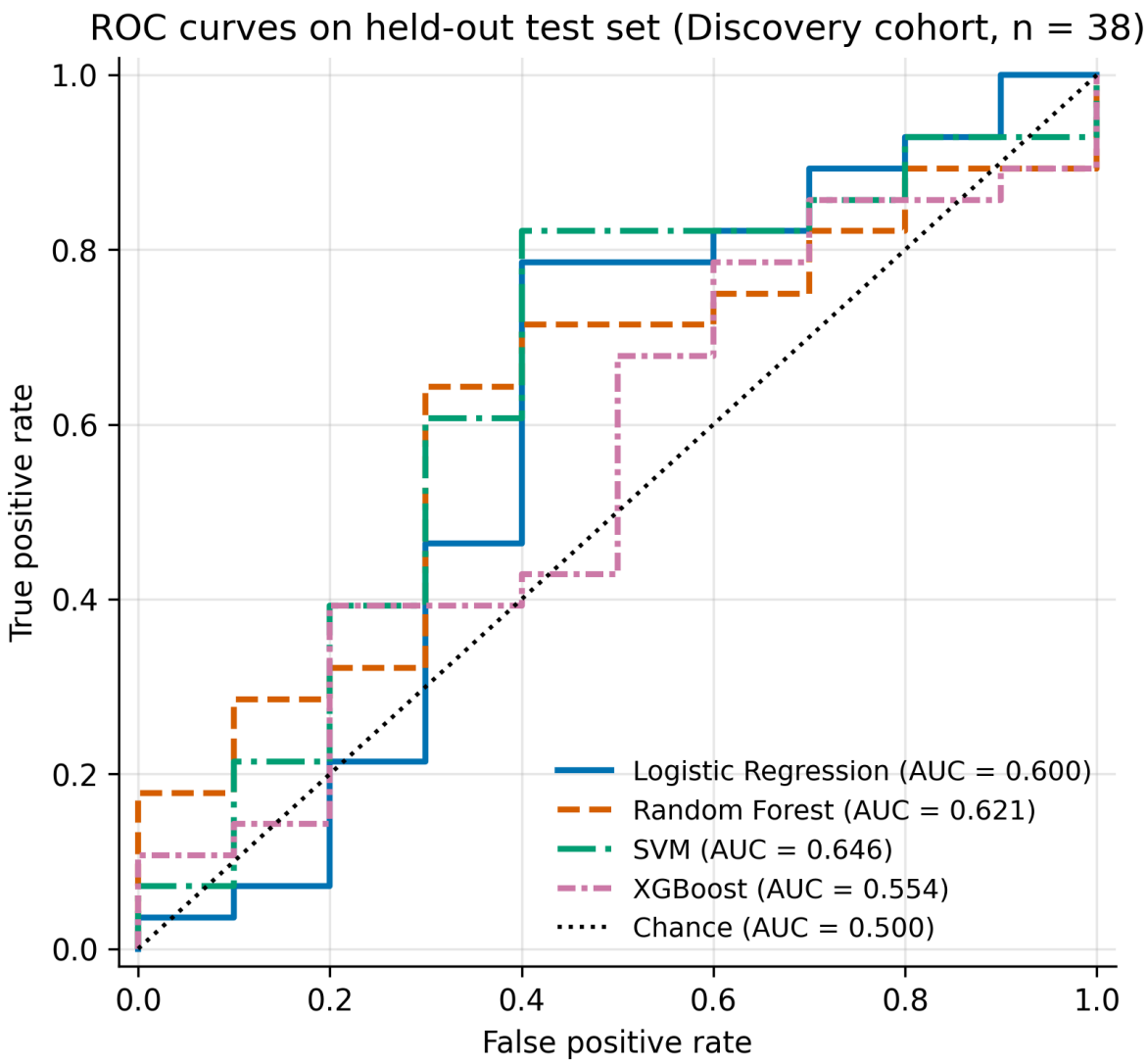


Figure 3: Receiver operating characteristic curves for all four tuned models on the held-out test set (n=38).

Test-set ROC-AUC values range from 0.554 (XGBoost) to 0.646 (SVM), against a chance baseline of 0.500. Figure 3 presents the ROC curves for all four models, demonstrating that each curve tracked closely along the diagonal. This pattern indicates that the models could not separate the two risk classes at any decision threshold. It is not just that the default threshold was poorly calibrated, but that no threshold existed at which the model's sensitivity and specificity both exceeded chance expectations.

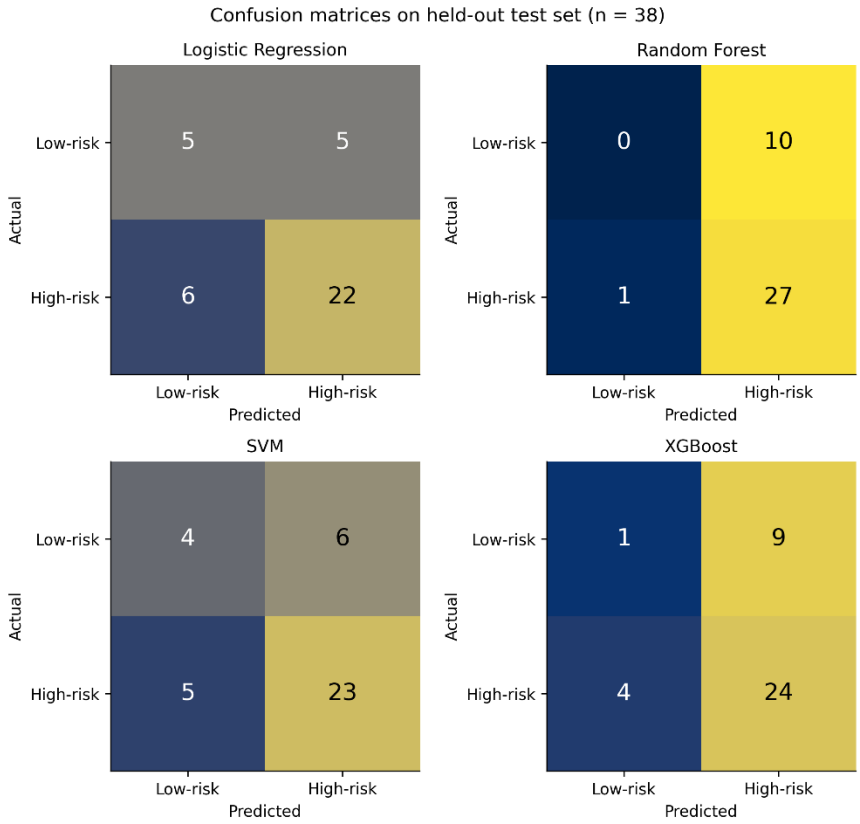


Figure 4: Confusion Matrices for all four tuned models on the held-out test set (n=38). Rows represent actual class membership, and columns represent predicted class membership. Random Forest Classified all ten low-risk individuals as high-risk while Logistic Regression achieved the highest low-risk recall at 5 of 10.

The confusion matrices, presented in Figure 4, revealed a striking pattern that aggregate metrics obscured. Random Forest achieved the joint-highest overall accuracy at 0.711 while correctly identifying none of the ten low-risk individuals in the test set. Its confusion matrix showed zero true negatives: every low-risk sample was misclassified as high-risk. The model's apparent accuracy was therefore attributable entirely to the majority class. It had learned to predict high-risk for every input, which is the trivial strategy that any clinician could perform without a model. For reference, a naive classifier assigning every sample to the high-risk group would achieve 0.737 accuracy on this test set (28 of 38 samples are high-risk), which exceeds the accuracy of all four tuned models.

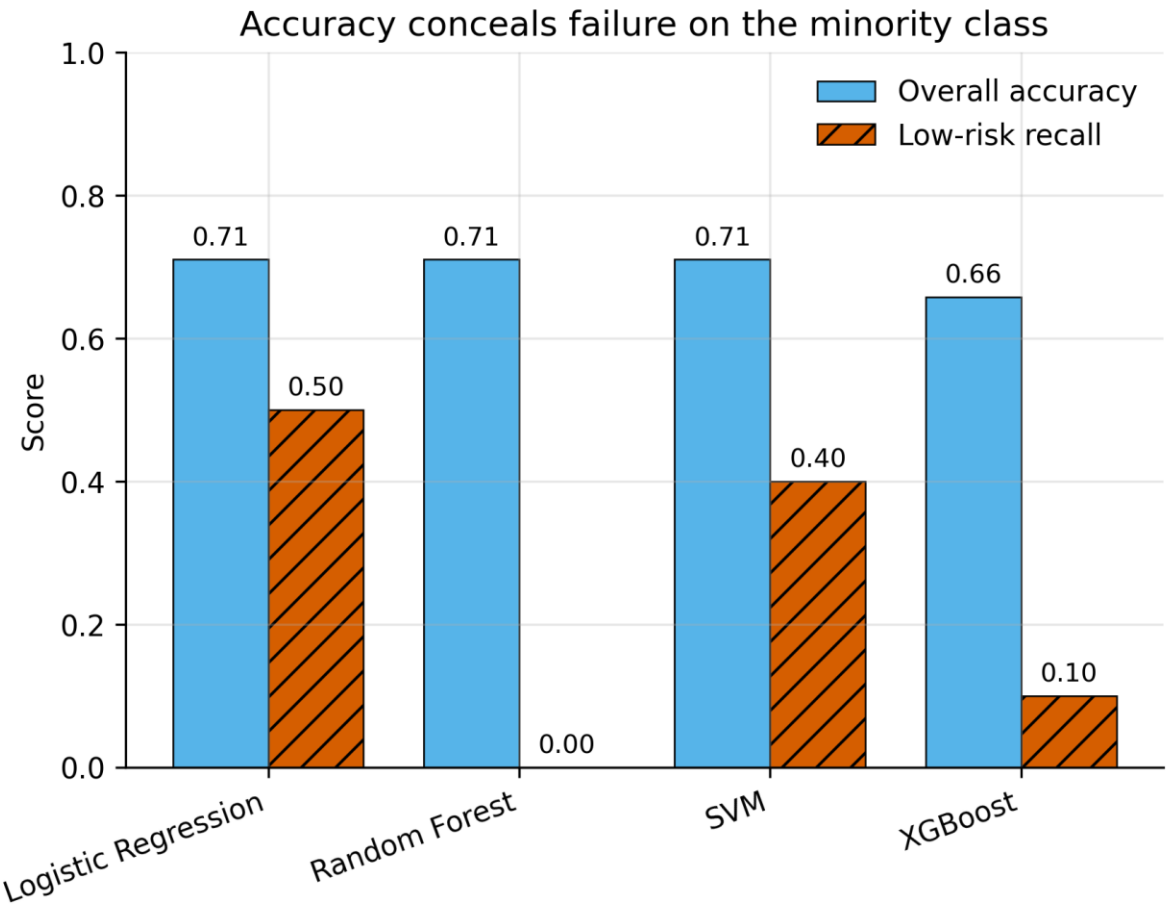


Figure 5: Overall accuracy versus low-risk recall for all four tuned models. The divergence between the two metrics reveals that accuracy is actively misleading in this imbalanced setting: Random Forest achieves 0.71 accuracy with 0.00 low-risk recall, indicating it classified every sample as high-risk.

Figure 5 made this accuracy-recall inversion visible. The overall accuracy bars appeared respectable across all four models, while the low-risk recall bars collapsed.

Logistic Regression achieved the best low-risk recall, correctly identifying five of ten low-risk individuals (recall 0.50) while SVM identified four of ten (recall 4.0). However, even these results should not be over interpreted: the difference between identifying five low-risk individuals and zero rests on the classification of handful of cases out of ten. With a test set containing only 10 low risk samples, the confidence interval around any per-class recall estimate is extremely wide, and the apparent ranking of models should be treated as indicative rather than definitive.

SVM achieved the highest ROC-AUC among the four models at 0.646, though this value remained within plausible chance variation for a 38-sample test set. XGBoost identified only one low-risk individual (recall 0.10) and achieved the lowest ROC-AUC at 0.554. The model ranking differed depending on which metric was privileged: SVM ranked first on ROC-AUC but shared the same overall accuracy as Logistic

Regression and Random Forest (0.711), while Logistic Regression ranked first on low-risk recall but second on ROC-AUC performance (0.600). This instability in the model ranking is itself a reflection of near-chance performance, in which all models are performing close to random and small stochastic variations determine which appears the best under any given criterion.

4.4 Effect of the Platform Confound on Model Discrimination

The practical significance of the platform confounds identified in section 4.1 was quantified by comparing the held-out test set ROC-AUC achieved when models were trained on all 333 samples (with the platform confound present) against the ROC-AUC achieved after restriction to the 187 Discovery cohort samples (confound removed). Table 9 represents this comparison.

Model	Test AUC — All 333 (Confounded)	Test AUC — Discovery Only	Δ
Logistic Regression	0.744	0.600	−0.144
Random Forest	0.721	0.621	−0.100
SVM	0.779	0.646	−0.133
XGBoost	0.777	0.554	−0.223

Table 9: Effect of removing the platform confound on held-out discrimination. Restricting the analysis to a single array platform reduced mean test ROC-AUC by 0.150, from 0.755 to 0.605, approaching the chance threshold consistently across all four algorithms.

As seen in Table 9, All four models exhibited a substantial decline in test-set ROC-AUC following removal of the platform confound. The mean AUC across algorithms fell from 0.755 under the confounded design to 0.605 under the Discovery-only design, a reduction of 0.150. XGBoost displayed the largest individual decline (Δ = −0.223), falling from the strongest confounded performance (0.777) to the weakest confound-free result (0.554); this is consistent with gradient boosting's capacity to exploit subtle distributional artefacts such as batch-level technical signatures. The remaining three models converged within a narrow range (0.600–0.646), with none achieving discrimination substantially above chance (0.5).

These results indicate that a meaningful proportion of the predictive signal available in the full 333-sample dataset was attributable to systematic differences between the EPIC v1 and v2 array platforms rather than to biologically informative methylation variation. The confounded AUC values (0.721–0.779) would, if reported in isolation, suggest moderate discriminative ability; their collapse upon platform restriction demonstrates that this interpretation would have been artefactual. This finding reinforces

the decision to restrict all primary analyses to the Discovery cohort, accepting reduced statistical power in exchange for validity.

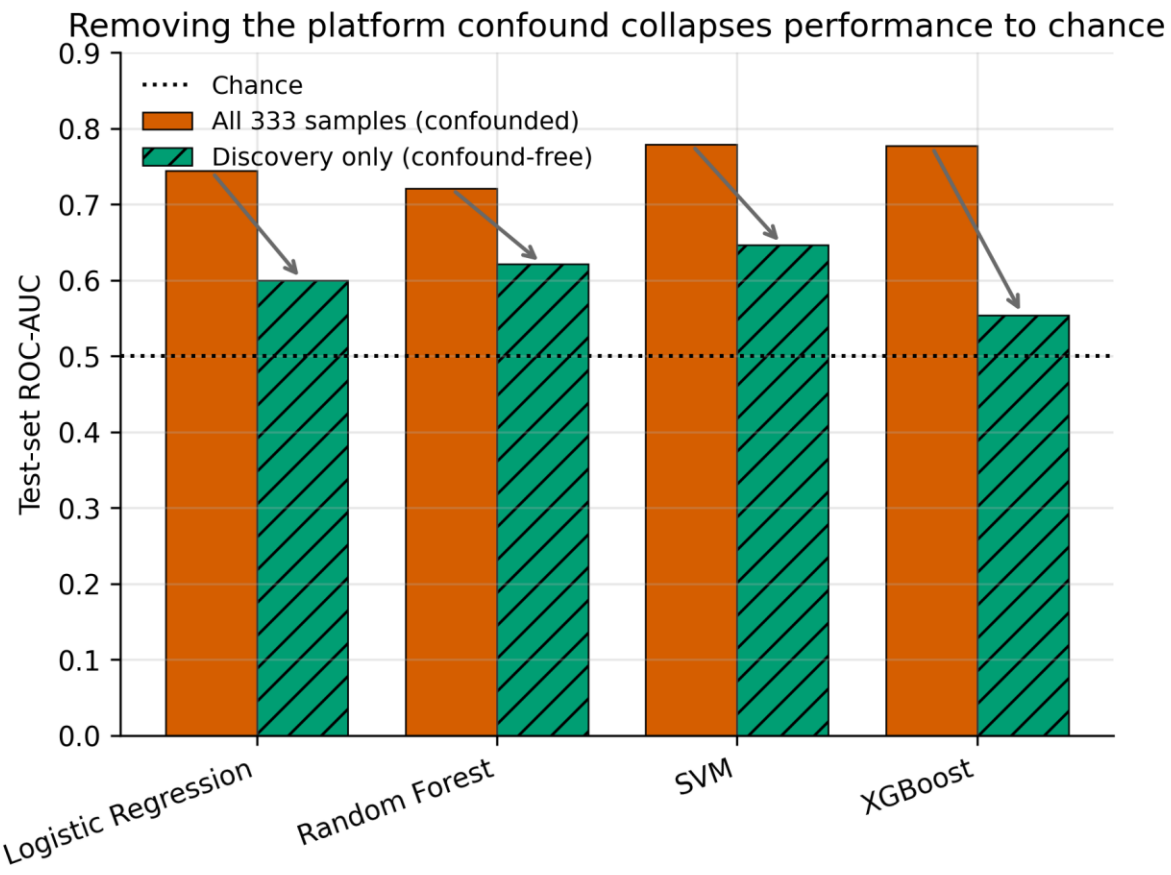


Figure 6: The confound removal collapses performance to chance. Removing the platform confound by constricting analysis to the Discovery cohort (single array platform) reduced test-set ROC-AUC by 0.100 to 0.223 across all four models, from a mean of 0.755 to 0.605. The consistency of this collapse across algorithms of markedly different inductive bias indicated that the effect is a property of the data, not of any particular model.

Across all four algorithms, removing the platform confound reduced held-out discrimination by between 0.100 and 0.223 AUC, to a mean of 0.605, approaching the 0.500 chance threshold. Figure 6 presents this result visually, with arrows indicating the direction and magnitude of the performance drop for each model. The consistency of this effect is its most important evidential property. Logistic Regression is Linear; Random Forest is an ensemble on non-linear decision trees; SVM operates in a transformed kernel space; XGBoost sequentially corrects residual errors. These four algorithms learn in fundamentally different ways and exploit different types of structure found within the data. The fact that all four models experienced a similar collapse in discrimination upon confound removal indicates that the effect reflects a property of the data, specifically, the loss of the platform-correlated shortcut, rather than a sensitivity particular to any one algorithmic approach.

The interpretation is clear. The apparent predictive performance obtained when training on the full dataset was greatly attributable to the technical signature of array platform rather than any signal of biological origin associated with prediabetes risk. Once this shortcut was removed, no model achieved discrimination meaningfully above chance, regardless of its complexity or inductive bias. This finding validates the decision to restrict the analysis to the Discovery cohort: had the restriction not been

applied, the study would have reported apparently promising AUC values in range of 0.72 to 0.78, leading to conclusion about methylation-based risk prediction that were, in fact, just conclusion about platform detection.

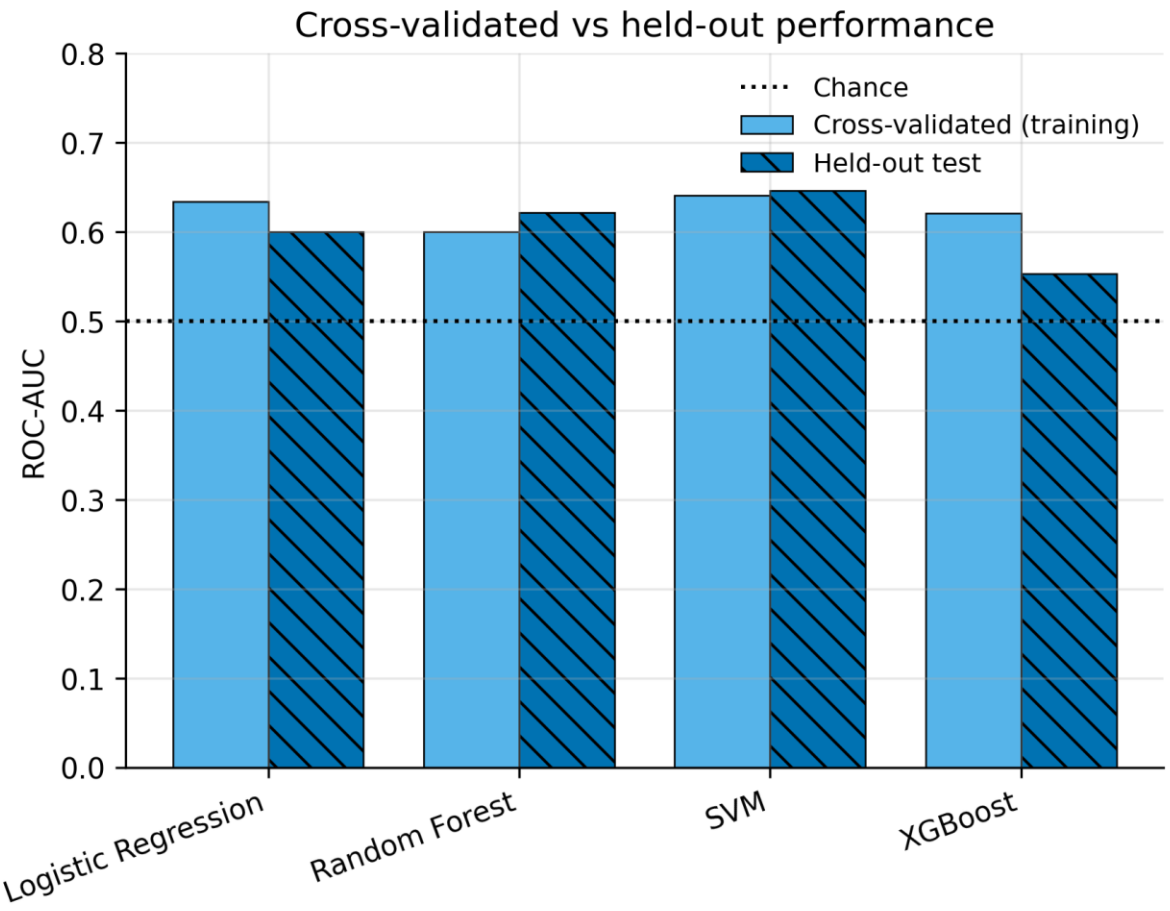


Figure 7: Cross validated ROC-AUC (tuned models, training set) versus held-out-test-set ROC-AUC. The optimism gap between training-set-cross-validation and test-set evaluation is modest and consistent, and both sets of values sit close to the chance baseline.

Figure 6 illustrates two features of the results that are diagnostically informative. First, the gap between cross-validated and held-out performance is notably small and, in the cases of Random Forest and SVM, the test-set AUC marginally exceeds the cross-validated estimate. In a typical machine learning study, a modest optimism gap is expected because cross-validation estimates are computed on data that the tuning process has searched over. The near-absence of this gap here is consistent with models that have found no exploitable structure during training: there is nothing to overfit to, so there is no inflation to lose at test time. Second, every bar in the figure, whether cross-validated or held-out, sits within a narrow margin just above the 0.500 chance line. The clustering of all eight values between 0.554 and

0.646 reinforces the conclusion drawn from the individual metrics: no algorithm, regardless of its complexity or inductive bias, achieved discrimination that could be considered clinically or statistically meaningful. The figure provides a concise visual summary of the central negative result — that the methylation features available after confound removal do not support reliable prediabetes risk classification at this sample size.

4.5 Model Interpretability and Biomarker Identification

The fifth research objective specified the identification of biologically relevant CpG sites contributing to prediabetes risk using explainable artificial intelligence methods. As described in Section 3.8, the study design incorporated an explicit precondition: feature attribution analysis would be pursued only if the models achieved discrimination meaningfully above chance on the held-out test set.

No model satisfied this condition. The highest test-set ROC-AUC achieved was 0.646 (SVM), a value that, at a sample size of 38, falls within the range attributable to chance variation around the 0.500 null hypothesis. Feature attribution analysis was therefore not pursued. SHAP values derived from a classifier that does not reliably separate the outcome classes would describe the internal weighting of a near-random model rather than any biological signal and presenting such attributions as candidate biomarkers would constitute scientific over-claim. The decision not to generate and report feature attributions from chance-level models is itself a methodological contribution: it demonstrates the application of interpretive restraint in a domain where the temptation to construct post-hoc biological narratives from model outputs is considerable, particularly when the underlying data contains hundreds of thousands of potential features from which superficially plausible stories could be assembled.

Chapter 5: Discussion

5.1 Summary of Key Findings

This study was created to evaluate the performance of four supervised machine learning algorithms for classifying prediabetes risk from genome-wide DNA methylation data. Three principal findings emerged. First, the dataset contained a previously uncharacterised structural confound in which array platform version, study cohort, and outcome class were perfectly correlated, such that platform alone could predict risk group with complete accuracy. Second, upon removal of this confound, no model achieved discrimination meaningfully above chance on the held-out test set, with ROC-AUC values ranging from 0.554 to 0.646 against a 0.500 baseline. Third, accuracy proved to be an actively misleading metric in this imbalanced setting, with one model (Random Forest) achieving 71.1% accuracy while correctly identifying none of the ten low-risk individuals in the test set. These findings are discussed further below.

5.2 The Platform Confound as a Methodological Contribution

The identification of the platform confound represents the most consequential finding of this study, both for its implications for the present analysis and as a methodological contribution to the broader field. Batch effects arising from differences in array platform, processing laboratory, or sample handling conditions are a well-documented source of misleading associations in genomic studies. Research by Leek et al. (2010) demonstrated that unaccounted batch effects are among the most common source of irreproducible findings in high-throughput biology, and subsequent work has repeatedly shown that machine learning classifiers trained on confound data learn the batch signature rather than the biological signal of interest. The confound in the GSE315764 was particularly severe because it was perfect: every Replication sample was assayed on EPIC v2 and belonged to a high-risk cluster, while every low-risk sample was assayed on EPIC v1 within the Discovery cohort. A model had no need to learn any methylation biology to achieve strong discrimination; it needed only to detect which array version had generated the data. The supporting evidence from the variance filter precomputation strengthened this interpretation: 95,082 CpG sites that had appeared biologically variable across the full dataset lost their variability when measured with a single platform, confirming that their apparent variation was platform-driven rather than individual-driven.

This finding has implications beyond the specific data used here. It demonstrates that standard performance metrics, applied naively, can report genuinely impressive-looking numbers that reflect technical artefact rather than a biological discovery. The mean confounded AUC of 0.755 across four machine learning algorithms would, in many published studies, be reported as evidence that DNA methylation can predict prediabetes risk. Only by recognising and accounting for the platform confound did the study reveal that the honest estimate of predictive performance was 0.605; approaching chance. The methodological lesson is that confound detection should precede not follow, model evaluation, and that the absence of confound documentation in a dataset does not constitute evidence that no confound exists.

5.3 Performance in Context: Comparison with Existing Literature

The near-chance performance observed after confound removal is consistent with several features of the existing literature. Lai et al. (2025) identified CpG sites longitudinally associated with T2D progression but explicitly noted that prediabetes-specific associations were limited and required validation in larger cohorts. Willmer et al., (2018) concluded that while individual CpG sites show statistically significant associations with T2D-related phenotypes, the effect sizes are typically small and the translational pathway to validated biomarkers remains underdeveloped. The results of the present study are compatible with this picture: individual methylation differences between prediabetes risk groups may exist but may be too subtle for reliable detection with 187 samples.

Several contextual factors may contribute to the observed results. The high-risk clusters (3,5 and 6) represent three distinct metabolic profiles that were collapsed into a single class for this analysis. If the methylation signatures of these three clusters differ from one another as much as they differ from the low-risk cluster, the high-risk class would be heterogenous, diluting any consistent or contrastive signal. In addition, the original cluster assignments were derived from metabolic profiling; methylation was measured in peripheral blood rather than in metabolically active tissues such as pancreatic islets or adipose tissue, introducing an additional layer of biological distance between the predictor variables, and the metabolic phenotype being predicted.

Lalani et al., (2025) conducted a scoping review of machine learning applications in prediabetes and identified a general lack of systematic comparison between algorithms applied to this specific prediction task, and model interpretability. The present study addresses that gap directly, and its finding, that four diverse algorithms perform equivalently at chance level is itself informative. It suggests that the barrier to prediction in this setting is not algorithmic, but data driven: no model can extract a signal that the data does not contain at a detectable strength.

5.4 The Accuracy-Recall Inversion and Metric Selection

The second major finding of this study concerns the relationship between overall accuracy and minority-class recall in imbalanced classification settings. Random Forest achieved 71.1% overall accuracy on the held-out test set, a figure that, in isolation, might appear adequate for a clinical screening tool. However, this accuracy was achieved by classifying every single sample as high-risk, correctly predicting the majority class while providing no stratification whatsoever. A naïve classifier assigning every individual to the high-risk group would achieve 73.7% accuracy on the same test set, outperforming all four tuned models.

This result illustrated a broader principle that is insufficiently emphasised in many applied machine learning studies: in imbalanced clinical datasets, accuracy is not merely uninformative but misleading (Ghanem et al, 2023). It rewards the behaviour least useful to the clinician, correctly identifying individuals who were already presumed to be at risk, whilst ignoring the behaviour most useful,

identifying the minority of individuals whose risk status differs from the base rate. The F1 score, which is often proposed as a correction for imbalanced settings, proved only partially effective in this case, as it was computed for high-risk (majority) class and therefore still rewarded majority-class prediction.

The clinical implication is that metric selection in prediabetes screening studies must be aligned with the clinical question being asked. If the purpose of a model is to identify who can safely be excluded from an intense follow up (to identify low risk samples for example), then low-risk recall is the metric of interest and a model with 70% accuracy and 0% low-risk recall is clinically worthless. The ROC-AUC, as a threshold -independent measure of overall discriminative ability, provided the most honest aggregate assessment of model performance in this study, and its proximity to chance confirmed what the confusion matrices showed at the individual-prediction level.

5.5 Implications for Clinical Translation

The results of this study do not support the clinical deployment of any of the evaluated models for prediabetes risk classification using DNA methylation data from the GSE315764 dataset. However, this negative conclusion should not be interpreted as evidence that DNA methylation is uninformative for prediabetes risk prediction in general. The result is specific to this dataset, sample size, tissue type (peripheral blood), cluster definitions, and this experimental design. A study with a larger, single platform cohort, tissue samples from metabolically active organs, and a more granular definition of prediabetes progression might yield fundamentally different results.

What the results do demonstrate clearly is the importance of rigorous practice in the development of any clinical prediction tool. Had the platform confound not been identified and accounted for, this study would have reported AUC values around 0.75 and might have concluded that machine learning models show promising performance for methylation-based prediabetes screening. Such a conclusion would have been incorrect, and any downstream clinical application built upon it would have inherited a performance estimate inflated by technical artefact. The methodological contribution of this study lies in demonstrating that the path from data to conclusion must include systematic checks for confounds, honest reporting of null results, and resistance to the interpretive temptation to extract biological narratives from chance-level models.

5.6 Limitations

Several limitations of this study should be acknowledged. The most fundamental is sample size. After restriction to the discovery cohort, only 187 samples were available for the full analysis, and the held-out test set contained only 38 samples, including just 10 low-risk individuals. This sample size is small by the standards of machine learning and limits both the statistical power to detect weak signals and the precision of performance estimates. The wide standard deviations observed in cross-validated metrics (up to 0.159 for ROC-AUC) reflect this limitation clearly.

The binary collapsing of four clinically distinct clusters into two risk groups may have obscured meaningful heterogeneity within the high-risk class. A multi-class analysis distinguishing clusters 3, 5 and 6 individually might reveal cluster-specific methylation patterns that are diluted in the binary formulation, though the small per-cluster sample sizes would pose additional statistical challenges. The study used methylation beta values as supplied by the dataset without performing additional quality control steps such as the removal of cross-reactive probes, sex chromosome probes or probes mapping to known polymorphic sites. While the variance filter likely removed many such probes, the absence of probe-level quality filtering represents a deviation from best practice in epigenome-wide analysis and may have introduced noise into the feature set.

The confound comparison presented in Section 4.4 used hardcoded AUC values from a prior full dataset run that is no longer reproduced by the current analytical pipeline. While the pipeline is deterministic (seeded), any modification to the seed, threshold, or pipeline structure would require rerunning the full-dataset analysis to regenerate these comparison values. This dependency on a prior run reduced the strict reproducibility of the confound comparison figure, though the analytical logic remains valid.

The inability to pursue SHAP-based feature attribution analysis, due to the near-chance discrimination achieved by all four models, represents a further limitation. While the decision not to report attributions from chance-level classifiers was methodologically appropriate, it meant that the fifth research objective, identifying the biologically relevant CpG sites contributing to prediabetes risk, could not be fulfilled. Future studies achieving meaningful discrimination would be positioned to apply SHAP analysis and identify candidate methylation biomarkers for further biological investigation.

Finally, this study examined a single dataset from a single research group. The findings regarding the platform confound may be specific to the design of GSE315764 and should not be assumed to generalise all methylation datasets. However, the methodological principle, that batch effects should be checked and accounted for before model evaluation is universal.

5.7 Future Work

Several directions for future research emerge from this study. First, the analysis should be replicated on larger, single platform methylation datasets with balanced representation of prediabetes risk groups. Datasets that have several hundred low-risk and several hundred high-risk individuals assayed on the same platform would provide the statistical power that the present study lacked, enabling more precise estimation of whether a methylation signal exists and, if so, how strong that signal is.

Second, multi-class classification distinguishing the individual high-risk clusters (3,5 and 6) could reveal cluster-specific methylation patterns that are obscured by the binary formulation. This would require larger per-cluster sample sizes than are available in the GSE315764 but could provide biologically richer insights into the epigenetic heterogeneity of prediabetes.

Third, the integration of methylation data with other omics modalities, such as transcriptomic, metabolomic, or proteomic data, may improve predictive performance by capturing more disease

relevant information across multiple biological layers. Rönn et al. (2024) demonstrated the value of muti-omics integration for T2D prediction in pancreatic islet tissue, and extending such approaches to blood-based screening represents a natural progression.

Fourth, deep learning approaches such as convolutional neural networks or attention-based architectures may be able to detect spatial patterns in methylation data that the classical algorithms evaluated here cannot capture. However, such approaches typically require substantially larger sample sizes and sacrifice the interpretability that this study prioritised, and their application would need to be carefully justifies against the risk of overfitting.

Fifth, the confound detection methodology developed in this study, metadata cross tabulation, platform-stratified variance analysis, and within-platform versus cross-platform performance comparison, could be formalised into a reusable quality assurance framework for genomic machine learning studies. Such a framework would help researchers identify analogous confounds in other datasets before committing resources to model development, potentially preventing the publication of inaccurate, confound-driven results.

Chapter 6: Conclusion

This study evaluated the performance of four supervised machine learning algorithms (Logistic Regression. Random Forest, Support Vector machine, and XGBoost) for classifying prediabetes risk using genome-wide methylation data from the GSE315764 dataset. The study addressed five research objectives: data preprocessing, feature selection, model training, performance comparison, and biomarker identification.

The most consequential finding was the identification of a structural confound within the dataset: array platform version, study cohort, and outcome class were perfectly correlated, such that models trained across both cohorts could achieve apparently strong discrimination (mean AUC 0.755) by detecting the technical signature of the array rather than any biological methylation signal. Upon restriction to a single platform subset to eliminate this confound, all four models produced held-out test set ROC-AUC values in the range of 0.554 to 0.646, approaching the 0.500 chance threshold. The consistency of this near-chance performance across four algorithms of markedly different inductive bias confirmed that the results reflected a property of the data rather than a limitation of any particular modelling approach.

A secondary finding demonstrated that overall accuracy was actively misleading in this imbalanced setting: Random Forest achieved 71.1% accuracy while correctly identifying none of the ten low-risk test individuals, performing worse than a naive classifier that assigns every sample to the majority class. This result highlights the importance of selecting evaluation metrics aligned with the clinical question being asked, and of inspecting confusion matrices rather than relying on aggregate scores.

Given the absence of meaningful held-out discrimination, feature attribution analysis via SHAP was not pursued. This decision reflects a principled commitment to interpretive restraint: attributions from chance-level models described noise, not biology, and reporting them as candidate biomarkers would constitute over-interpretation.

The study contributes to the field in three ways. First, it provides a rigorous, methodologically transparent evaluation of machine learning for methylation-based prediabetes risk prediction, demonstrating that apparent predictive performance can be substantially inflated by undetected batch effects.

Second, it offers a case study in responsible analytical practices; confound detection, leakage prevention, honest reporting of null results, and appropriate metric selection, that is applicable to any genomic machine learning study. Third, it demonstrates that a negative results, when reached through sound methodology, is at least as informative as a positive one: it delimits what the data can and cannot support, identifies the structural barriers to prediction, and directs future work towards the specific improvements (larger samples, single platform designs, multi-omics integration) most likely to alter the outcome.

The absence of predictive signal in this dataset should not discourage further research into methylation-based prediabetes screening. It should, however, encourage researchers to subject their datasets to the same degree of structural scrutiny applied here before drawing conclusions about predictive

performance, and to recognise that methodological rigour is not an obstacle to discovery but its prerequisite.

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Appendix

Appendix A: Ethics Approval Letter

Ethical approval for this project was granted by the University of Sheffield (departmental self-declaration, approved 6 July 2026).

The approval letter is omitted from this public copy because it contains personal administrative details.

Appendix B: Supplementary Background on Diabetes Mellitus

B.1 Global Epidemiology of Diabetes

The International Diabetes Federation (IDF) Diabetes Atlas, 10th edition, estimated that 537 million adults aged 20–79 years were living with diabetes worldwide in 2021, representing a global prevalence of 10.5%. This figure is projected to rise to 643 million by 2030 and 783 million by 2045, driven by population growth, ageing, urbanisation, and increasing rates of obesity and physical inactivity (Sun et al., 2022). Type 2 diabetes mellitus accounts for over 90% of all diabetes cases globally (Magliano & Boyko, 2021). The IDF further estimated that global diabetes-related health expenditure reached 966 billion USD in 2021 (Sun et al., 2022).

A significant proportion of diabetes remains undiagnosed. The IDF estimated that nearly 45% of all adults with diabetes in 2021, were unaware of their condition (Magliano & Boyko, 2021). This diagnostic gap is particularly pronounced in low-and middle-income countries, where access to screening and healthcare infrastructure is limited. The implications for public health are substantial: undiagnosed diabetes delays treatment initiation and increases the risk of complications including cardiovascular disease, nephropathy, retinopathy, and peripheral neuropathy (Chatterjee, 2017)

B.2 Pathophysiology of Type 2 Diabetes

Type 2 diabetes mellitus is a progressive metabolic disorder characterised by two interrelated defects: insulin resistance in peripheral tissues (primarily skeletal muscle, liver, and adipose tissue) and a progressive decline in pancreatic β-cell function leading to inadequate insulin secretion (DeFronzo et al., 2015). In the early stages, β-cells compensate for peripheral insulin resistance by increasing insulin output, maintaining near-normal glucose levels. As β-cell function deteriorates over time, however, this compensatory mechanism fails, and hyperglycaemia develops with up to 50% of cell loss at diagnosis (Chatterjee et al., 2017).

The progression from normal glucose tolerance through impaired glucose tolerance (prediabetes) to clinical diabetes typically spans several years and is influenced by both genetic susceptibility and environmental factors. Genome-wide association studies have identified over 240 genetic loci associated with T2D risk, though the cumulative effect of common variants explains only a fraction of disease heritability (Mahajan et al., 2018). Environmental and lifestyle factors, including dietary composition, physical activity levels, adiposity, and chronic stress, interact with genetic predisposition through epigenetic mechanisms including DNA methylation, thereby modulating disease risk across the lifespan (Ling & Rönn, 2019).

B.3 Prediabetes: Definitions and Diagnostic Criteria

Prediabetes is defined as a metabolic state in which glycaemic parameters are elevated above normal but remain below the diagnostic threshold for diabetes. The World Health Organization (WHO) and the American Diabetes Association (ADA) use slightly different criteria for its identification. The WHO

defines impaired fasting glucose (IFG) as a fasting plasma glucose of 6.1–6.9 mmol/L and impaired glucose tolerance (IGT) as a 2-hour plasma glucose of 7.8–11.0 mmol/L following a 75g oral glucose tolerance test. The ADA uses a lower IFG threshold of 5.6–6.9 mmol/L and additionally recognises HbA1c values of 39–47 mmol/mol (5.7–6.4%) as indicative of prediabetes (American Diabetes Association, 2023).

The clinical significance of prediabetes lies in its predictive value for future T2D development. A meta-analysis by Tabak et al. (2012) established that 5–10% of individuals with prediabetes progress to clinical diabetes annually, and that without intervention, the majority will develop T2D within 10 years. However, this progression is modifiable: landmark clinical trials including the Diabetes Prevention Program (DPP) demonstrated that intensive lifestyle intervention reduced the incidence of T2D by 58% over a three-year period in high-risk individuals (Knowler et al., 2002).

Appendix C: DNA Methylation

C.1 The Molecular Mechanism of DNA Methylation

DNA methylation is an epigenetic modification involving the covalent addition of a methyl group ($–CH_3$) to the 5-carbon position of cytosine residues in DNA, forming 5-methylcytosine (5mC). This reaction is catalysed by a family of enzymes known as DNA methyltransferases (DNMTs). In mammals, DNMT3A and DNMT3B are responsible for establishing new (de novo) methylation patterns during embryonic development, while DNMT1 maintains existing methylation patterns through cell division by copying the methylation state from the parent strand to the newly synthesised daughter strand (Jones, 2012).

Methylation occurs predominantly at cytosine–phosphate–guanine (CpG) dinucleotide sites. In the mammalian genome, approximately 70–80% of CpG sites are methylated. However, regions of unusually high CpG density, termed CpG islands (CGIs), are typically hypomethylated. Approximately 70% of annotated gene promoters are associated with a CpG island (Deaton & Bird, 2011). When CpG islands in promoter regions become methylated, gene transcription is typically repressed through two complementary mechanisms: direct steric inhibition of transcription factor binding to methylated DNA, and recruitment of methyl-CpG-binding domain (MBD) proteins, which in turn recruit histone deacetylases and chromatin remodelling complexes that establish a repressive chromatin state (Jones, 2012; Moore et al., 2013).

C.2 CpG Sites and CpG Islands

A CpG site is a specific location in the genome where a cytosine nucleotide is followed by a guanine nucleotide in the 5’ to 3’ direction, linked by a phosphodiester bond. In vertebrate genomes, CpG dinucleotides are globally depleted relative to their expected frequency due to the spontaneous deamination of 5-methylcytosine to thymine over time (Gardiner-Garden & Frommer, 1987; Moore et al., 2013).

CpG islands are defined as genomic regions of at least 200 base pairs with a GC content greater than 50% and an observed-to-expected CpG ratio greater than 0.6 (Gardiner-Garden & Frommer, 1987). The human genome contains approximately 28,000 CpG islands, of which roughly 70% overlap with gene promoter regions (Illingworth et al., 2010). The methylation status of CpG islands in promoter regions is strongly associated with transcriptional activity: unmethylated CpG islands are associated with active or poised gene expression, while methylated CpG islands are associated with stable long-term gene silencing, as observed in X-chromosome inactivation and genomic imprinting (Deaton & Bird, 2011).

C.3 The Illumina Infinium MethylationEPIC Platform

The Illumina Infinium MethylationEPIC BeadChip is a microarray-based platform for genome-wide DNA methylation profiling at single-CpG resolution. The EPIC array interrogates over 850,000 CpG sites per sample, retaining more than 90% of the probes from its predecessor, the HumanMethylation450

BeadChip, while adding over 350,000 CpGs at regions identified as potential enhancers by the FANTOM5 and ENCODE projects (Pidsley et al., 2016).

The assay workflow begins with bisulfite conversion of genomic DNA, a chemical treatment in which unmethylated cytosines are deaminated to uracil, while methylated cytosines (5mC) remain unchanged. This creates a sequence difference between methylated and unmethylated alleles at every CpG site. The bisulfite-converted DNA is then amplified, fragmented, and hybridised to the BeadChip, which contains locus-specific oligonucleotide probes attached to microscopic beads arranged in a microarray format (Zhou et al., 2017).

The EPIC array employs two probe chemistries. Infinium I probes use two bead types per CpG locus: one complementary to the methylated sequence and one to the unmethylated sequence, with single-base extension incorporating a labelled nucleotide. Infinium II probes use a single bead type that hybridises to the target regardless of methylation status, with the methylation state determined by which labelled nucleotide (associated with either a green or red fluorescent channel) is incorporated during single-base extension (Zhou et al., 2017). The methylation level at each CpG site is reported as a beta value, calculated as the ratio of the methylated signal intensity to the total signal intensity (methylated plus unmethylated), yielding a continuous value between 0 (no methylation) and 1 (complete methylation) (Pidsley et al., 2016).

Appendix D: Glossary of Key Terms

Beta value: A continuous measure of DNA methylation at a specific CpG site, calculated as the ratio of the methylated signal intensity to the total signal intensity (methylated plus unmethylated). Beta values range from 0 (completely unmethylated) to 1 (completely methylated) *(Pidsley et al., 2016).*

Biomarker: A defined characteristic that is measured as an indicator of normal biological processes, pathogenic processes, or responses to an exposure or intervention *(Mayeux, 2004).*

Bisulfite conversion: A chemical treatment of genomic DNA in which unmethylated cytosine residues are deaminated to uracil, while methylated cytosines (5-methylcytosine) remain unchanged. This creates a sequence difference that enables methylation status at individual CpG sites to be determined through subsequent sequencing or hybridisation-based assays *(Pidsley et al., 2016).*

Class imbalance: A condition in supervised classification in which the number of observations belonging to one class substantially exceeds the number belonging to another, potentially causing classifiers to favour the majority class and achieve misleadingly high accuracy despite failing to identify minority-class instances *(He & Garcia, 2009).*

Confound (confounding variable): A variable that influences both the independent and dependent variables in a study, creating a false impression of an association between them that does not reflect a genuine relationship *(Andrade, 2024).*

CpG dinucleotide: A two-nucleotide sequence in DNA in which a cytosine is immediately followed by a guanine in the 5' to 3' direction, linked by a phosphodiester bond. The cytosine within a CpG dinucleotide is the principal substrate for DNA methylation in mammalian genomes *(Gardiner-Garden & Frommer, 1987).*

CpG island: A genomic region characterised by a GC content exceeding 50% and an observed-to-expected CpG ratio greater than 0.6. Approximately 70% of human gene promoters are associated with a CpG island, and their methylation status is closely linked to transcriptional regulation *(Gardiner-Garden & Frommer, 1987; Deaton & Bird, 2011).*

Cross-validation: A resampling procedure used to evaluate machine learning models on limited data, in which the dataset is partitioned into complementary subsets so that the model is trained on one subset and validated on another, with the process repeated across multiple partitions to produce a more robust estimate of predictive performance *(Arlot & Celisse, 2010).*

DNA methylation: An epigenetic modification involving the covalent addition of a methyl group to the 5-carbon position of a cytosine residue in DNA, forming 5-methylcytosine. This modification can regulate gene expression without altering the underlying DNA sequence, typically by repressing transcription when occurring at gene promoter regions *(Moore et al., 2013).*

Epigenetics: The study of changes in gene function that are heritable through cell division but do not involve alterations to the underlying DNA sequence. Epigenetic mechanisms include DNA methylation, histone modification, and non-coding RNA regulation *(Bird, 2007).*

False discovery rate (FDR): The expected proportion of rejected null hypotheses (discoveries) that are false positives. FDR-controlling procedures, such as the Benjamini–Hochberg method, are used in multiple hypothesis testing to balance the detection of true effects against the risk of reporting false findings *(Benjamani & Hochberg, 2018).*

Feature selection: The process of selecting a subset of relevant predictor variables for use in model construction, with the aim of reducing dimensionality, improving model interpretability, and mitigating overfitting by removing uninformative or redundant features *(Zhuang et al., 2012).*

GridSearchCV: A function within the scikit-learn Python library that performs an exhaustive search over a specified grid of hyperparameter values, evaluating each combination using cross-validation and selecting the configuration that optimises a chosen performance metric *(Scikit-learn developers, 2026).*

HbA1c (glycated haemoglobin): A blood test that reflects average plasma glucose concentration over the preceding two to three months by measuring the proportion of haemoglobin molecules to which glucose has become covalently attached. HbA1c values of 39–47 mmol/mol (5.7–6.4%) are used as a diagnostic criterion for prediabetes under ADA guidelines *(American Diabetes Association, 2023).*

Hyperparameter: A configuration variable of a learning algorithm that is specified prior to the commencement of training and governs the training process itself, as distinct from model parameters, which are estimated from the training data during fitting *(Scikit-learn developers, 2026).*

Impaired fasting glucose (IFG): A prediabetic metabolic state defined by a fasting plasma glucose concentration of 6.1–6.9 mmol/L under WHO criteria, or 5.6–6.9 mmol/L under ADA criteria, indicating elevated but sub-diagnostic glycaemia *(American Diabetes Association, 2023).*

Impaired glucose tolerance (IGT): A prediabetic metabolic state defined by a plasma glucose concentration of 7.8–11.0 mmol/L measured two hours after administration of a 75g oral glucose tolerance test, indicating impaired but sub-diagnostic postprandial glucose disposal *(American Diabetes Association, 2023).*

LASSO regression: Least Absolute Shrinkage and Selection Operator. A regularised regression method that applies an L1 penalty to the absolute magnitude of regression coefficients, shrinking uninformative coefficients to exactly zero and thereby performing simultaneous variable selection and regularisation *(Tibshirani, 1996).*

Matthew's Correlation Coefficient (MCC): A metric for evaluating the quality of binary classifications that incorporates all four quadrants of the confusion matrix (true positives, true negatives, false positives, and false negatives). MCC returns values between −1 (complete misclassification) and +1 (perfect classification), with 0 indicating performance equivalent to random prediction. It is considered the most balanced single-number metric for imbalanced classification tasks *(Chicco & Jurman, 2020).*

Overfitting: A modelling error that occurs when a statistical or machine learning model captures noise or patterns specific to the training data rather than the underlying generalisable relationship, resulting in strong training performance but poor predictive accuracy on unseen data *(Subramanian & Simon, 2013).*

Prediabetes: A metabolic condition in which blood glucose levels are elevated above the normal range but remain below the threshold for a clinical diagnosis of diabetes mellitus. Prediabetes encompasses both impaired fasting glucose and impaired glucose tolerance, and confers a substantially increased risk of progression to type 2 diabetes *(Tabák et al., 2012).*

ROC-AUC (Area Under the Receiver Operating Characteristic Curve): A threshold-independent metric that quantifies a binary classifier's ability to discriminate between two classes across all possible decision thresholds. ROC-AUC ranges from 0.0 to 1.0, where 0.5 indicates discrimination no better than chance and 1.0 indicates perfect separation of the two classes *(Hanley & McNeil,1982).*

SHAP (Shapley Additive Explanations): A model-agnostic framework for interpreting the output of machine learning models, grounded in the Shapley value from cooperative game theory. SHAP assigns each feature a value representing its contribution to the prediction for a specific instance, enabling both local (per-prediction) and global (across-dataset) explanations of model behaviour *(Lundberg & Lee, 2017).*

Stratified sampling: A sampling technique in which a population is divided into distinct subgroups (strata) based on a shared characteristic, and samples are drawn from each stratum in proportion to its representation in the population, ensuring that the resulting sample preserves the distributional properties of the original dataset *(Singh & Mangat, 1996).*

Type 2 diabetes mellitus (T2D): A chronic metabolic disorder characterised by persistent hyperglycaemia resulting from progressive pancreatic β-cell dysfunction and peripheral insulin resistance. T2D accounts for approximately 90–95% of all diabetes cases globally *(Chatterjee et al., 2017).*

Research Diary & Reflection

Supervision Meeting Summaries

Meeting 1: Project Scoping and Model Selection

Work done before meeting:

- Completed initial literature review on DNA methylation and T2D.
- Identified GSE315764 as the primary dataset.

Key points discussed:

- Selection of four machine learning algorithms (Logistic Regression, Random Forest, SVM, XGBoost) was agreed upon, spanning the interpretability-performance spectrum.
- Discussed the importance of choosing models that offer different inductive biases to enable a meaningful comparison.
- Feature selection methods were discussed: agreed on a multi-stage approach using variance filtering, LASSO, and Random Forest importance.
- Supervisor introduced the concept of SHAP for model interpretability and its relevance to clinical applicability.

Goals ahead of next meeting:

Begin data preprocessing and implement the feature selection pipeline.

Draft the methodology section covering model selection and feature selection rationale.

Meeting 2: Methodology Review and Design Justifications

Work done since last meeting:

- Completed data preprocessing pipeline including variance filtering and train/test split.
- Drafted methodology sections 3.2–3.5.

Key points discussed:

- Supervisor emphasised the importance of justifying every design choice with either a citation or an explicit rationale, particularly the binary cluster grouping strategy.
- Discussed the need to reference Singh et al. (2026) to justify why cluster 2 was designated as low-risk and clusters 3, 5, and 6 as high-risk, based on their metabolic profiles.
- Supervisor noted that the hyperparameter search space should be presented explicitly in a table so readers can see exactly which values were explored.
- Discussed the importance of reporting the final number of features retained after all selection stages.

Goals ahead of next meeting:

Add cluster grouping justification to section 3.2.

Create hyperparameter search space table.

Run the full modelling pipeline and generate initial results.

Meeting 3: Results Review and Document Structure

Work done since last meeting:

- Completed model training, tuning, and test-set evaluation.
- Identified the platform confound in the dataset structure.
- Drafted results chapter including tables and figures.

Key points discussed:

- Supervisor recommended adding a visual pipeline diagram showing the entire methodology from data acquisition through evaluation, to help readers grasp the full analytical workflow at a glance.
- Noted that SHAP was discussed in the methodology (section 3.7) without being introduced in the literature review. Supervisor advised adding a subsection to the literature review covering explainability in healthcare ML.
- Discussed the need for concepts to be introduced before they are discussed in depth, following a logical narrative structure.
- Supervisor recommended including Matthew's Correlation Coefficient (MCC) as an evaluation metric, as it provides a more balanced assessment of classification quality on imbalanced datasets than accuracy or F1 alone.
- Ethics approval letter should be included in the appendix.

Goals ahead of Submission Deadline:

Add MCC to the evaluation code and update all results tables.

Add SHAP/explainability subsection to the literature review.

Create the methodology pipeline diagram.

Finalise the discussion and conclusion chapters.

Feedback Reflections

Reflection 1: Incorporating Matthew's Correlation Coefficient

Feedback received: *Supervisor recommended adding MCC to the evaluation metrics, noting that it is the most informative single metric for imbalanced binary classification and would strengthen the assessment of model performance.*

This feedback led to one of the most technically instructive experiences of the project. Adding MCC required modifying the Python modelling script to compute *matthews_corrcoef* from scikit-learn at two points: within the cross-validation results (using *cross_val_predict*) and on the held-out test set. However, after adding the MCC code, the results became non-deterministic across runs. The same script, with the same random seed, produced different numbers each time it was executed.

Diagnosing this issue required a systematic understanding of how NumPy's global random state operates. The *cross_val_predict* function refits all models across five folds, consuming random numbers from NumPy's global pool. The LASSO feature selector used *liblinear*, which also draws from this global pool. Before the MCC code was added, the sequence of draws was always identical, so results were deterministic by coincidence. The MCC code disrupted this sequence, causing the LASSO selector to receive different random numbers on subsequent runs, which cascaded into different feature selections and different final results.

The fix was to add *random_state=42* to every LogisticRegression instance inside the LASSO selectors, giving each its own private random stream independent of the global pool. This made the pipeline deterministic by design rather than by accident. The experience strengthened my understanding of reproducibility in computational research: a script that produces identical results is not necessarily deterministic if its reproducibility depends on the absence of future code additions. Pinning every random process with an explicit seed is a defensive practice I will carry into future work. It also deepened my understanding of how scikit-learn pipelines interact with NumPy's random number generation, a topic not covered in standard tutorials but essential for rigorous experimental practice.

Reflection 2: Justifying the Cluster Grouping Strategy

Feedback received: *Supervisor noted that the binary collapsing of four prediabetes clusters into low-risk and high-risk groups required explicit justification, either citing the original study's rationale or providing an independent clinical argument.*

Initially, the methodology simply stated that cluster 2 was coded as low-risk and clusters 3, 5, and 6 as high-risk, without explaining why. This was a significant gap: a reader unfamiliar with the original Singh et al. (2026) study would have no basis for evaluating whether the grouping was clinically meaningful or arbitrary. In response, I added a paragraph to section 3.2 explaining that Singh et al. characterised cluster 2 by a metabolically favourable profile with low insulin resistance, low hepatic fat, and preserved beta-cell function, whereas the remaining clusters each exhibited distinct but consistently elevated metabolic risk features. I also noted that the binary formulation maximised statistical power given the sample size and aligned with the clinical decision of interest, which is itself fundamentally binary: whether an individual requires heightened monitoring or not.

This change improved the dissertation by ensuring that every analytical decision could be traced to a documented rationale. The skill I developed was learning to distinguish between decisions that are self-evidently justified and those that require explicit defence. In hindsight, the cluster grouping was a consequential design choice that directly shaped the classification task, and leaving it unjustified would have undermined the analytical integrity of the entire study. This experience taught me that a well-designed methodology section should enable a reader to understand not only what was done, but why each decision was made over its alternatives.

Reflection 3: Introducing Concepts Before Discussing Them in Depth

Feedback received: *Supervisor identified that SHAP was first mentioned in the methodology (section 3.7) as a tool for feature attribution but had not been introduced or contextualised in the literature review. This meant the reader encountered a specific analytical technique without any prior discussion of why explainability matters in clinical ML.*

In response, I added a new subsection (2.4b: Explainability in Machine Learning for Healthcare) to the literature review, covering the growing emphasis on model interpretability in clinical applications, the theoretical basis of SHAP in cooperative game theory, and the distinction between local and global explanations. I also added a concluding paragraph to section 2.4 that explicitly linked the literature review to the model selection strategy and the incorporation of explainability as a design objective. These additions ensured that when SHAP appeared in the methodology, the reader already understood its purpose and theoretical grounding.

This feedback strengthened my understanding of how academic writing differs from technical documentation. In a script or a notebook, it is acceptable to introduce a tool at the point of use. In a dissertation, the reader expects a logical progression from broad context through specific justification to

implementation detail. Each concept should be motivated before it is deployed. The skill I developed was structuring a document so that the literature review does not merely survey existing work but actively builds the intellectual foundation for the decisions made in the methodology. This principle, that every methodological choice should be grounded in the literature, is one I will apply to future research writing.

Reflection 4: Creating a Visual Pipeline Diagram

Feedback received: *Supervisor recommended including a visual representation of the entire analytical pipeline, from data acquisition through evaluation, to provide readers with an at-a-glance understanding of the methodology.*

I created a colour-coded flowchart showing all nine stages of the pipeline: data acquisition, initial preprocessing, confound detection, cohort restriction, train/test split, feature selection, model training, evaluation, and interpretability assessment. The diagram used distinct colours for each phase (preprocessing, confound handling, feature selection, modelling, evaluation) and included a dashed bracket indicating which stages operated within the cross-validation loop, a detail that is critical to understanding how data leakage was prevented but is difficult to convey in prose alone.

This addition improved the dissertation by making the overall structure of the analysis immediately accessible to readers who might otherwise struggle to reconstruct the full workflow from the narrative text. The skill I developed was learning to identify which information is best communicated visually rather than verbally. The cross-validation boundary, for example, is a spatial concept (some stages are inside the loop, some are outside) that a diagram conveys in seconds, but a paragraph can only describe sequentially. Effective scientific communication requires matching the presentation format to the nature of the information, and this was a concrete example of that principle in practice.

Reflection 5: Documenting the Feature Selection Output

Feedback received: *Supervisor asked what the final number of dimensions was after all filtering and feature selection procedures, noting that this information was absent from the methodology.*

This was a straightforward but important gap. The methodology described each stage of the feature selection pipeline but never stated how many features remained at the end. I added a paragraph to section 3.4 explaining that the exact number varied across cross-validation folds because both LASSO and Random Forest selection stages were refitted within each fold to prevent data leakage. The input was 193,638 CpG sites after variance filtering; LASSO reduced this to several hundred, and the Random Forest importance stage further reduced the set to those scoring above the mean threshold. I explained that the variability in feature counts across folds was an expected consequence of the leakage-safe pipeline design rather than an oversight.

This change improved the dissertation by providing the reader with a concrete sense of the dimensionality reduction achieved, which is essential for evaluating whether the feature selection was effective. The skill I strengthened was anticipating what information a reader needs in order to evaluate a methodology. I had understood the feature counts from inspecting my code outputs but had not recognised that a reader without access to those outputs would find the omission conspicuous. Scientific writing must be self-contained: every claim and every process described should include enough quantitative detail for the reader to assess its adequacy independently.

AI declaration

The web version of Gemini processed the following prompts and requests explained below.

AI was used to suggest the restructuring of a previously drafted abstract paragraph. AI suggested it be split into the headings seen above (Backgrounds, Aims, Methods, Results, Conclusion). The previous draft was a paragraph with no breaks or sections. It was pasted into the chat box before this query was made.

The prompt used was: “Please offer suggestions about how this abstract paragraph can be restructured into multiple sections to match conventional formatting standards (Backgrounds, Aims, Methods, Results, Conclusion).”

AI was also used to suggest how to structure the Summary of Key Findings in 5.1. It was used to draft the order that information would be introduced, presenting the most important information first. It suggested the presentation of the platform confound first, followed by the AUC-ROC results after confound removal and finally the accuracy measurements, which proved to be misleading metrics.

The prompt used to achieve this was: “What should be presented first within the Summary of Key Findings section, and the overall order that the report results should be presented in?”

AI was used to spell check the document. A draft of the report was pasted into the chat box before a prompt was created and sent to the AI model asking it to spell check the document.

The prompt used to do this was: “Please identify any spelling mistakes within this document and list them so that I can correct them.”